Assignment 1 – Computer Forensics

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Q1. Data Acquisition

a. Disk Image Creation

Step 1:

Select the “Create Disk Image…” within the “File” tab to start the process.

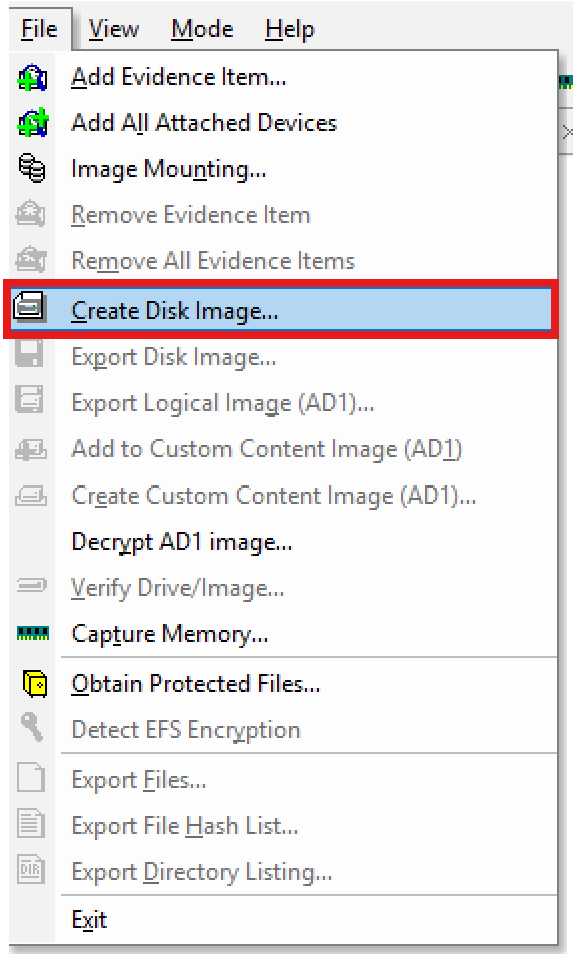


Figure 1: Creating new Disk Image

Step 2:

Select the “Physical Drive” option since we are creating a disk image from a physical USB drive.

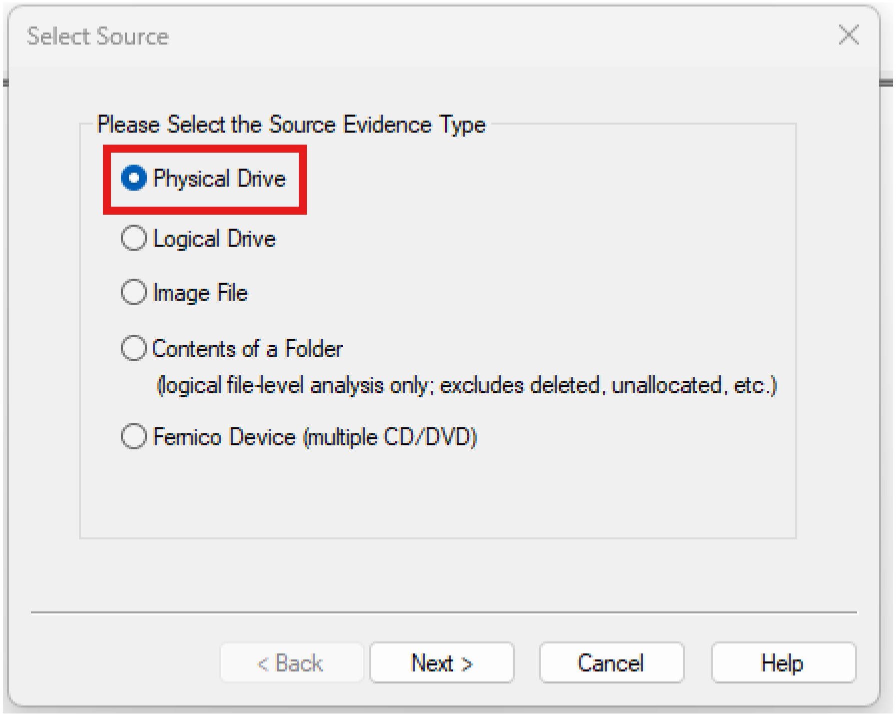


Figure 2: Selecting source

Step 3:

Selecting the appropriate drive for image creation.

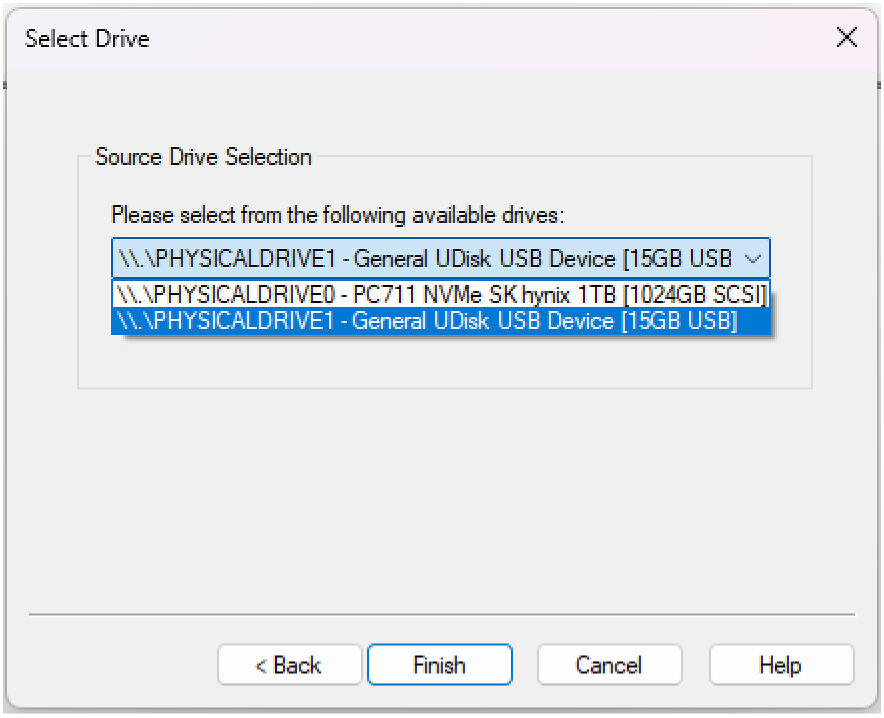


Figure 3: Selecting drive

Step 4:

Click add to configure the details of the drive image.

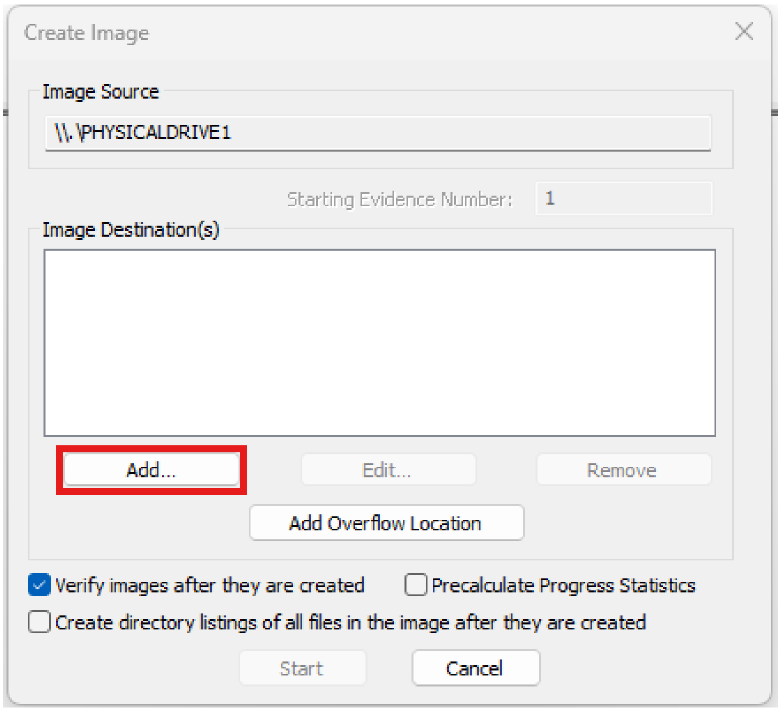


Figure 4: Configuring Image options

Step 5:

Select the type of the output image. We are selecting the “Raw (dd)” option, as requested within the assignment.

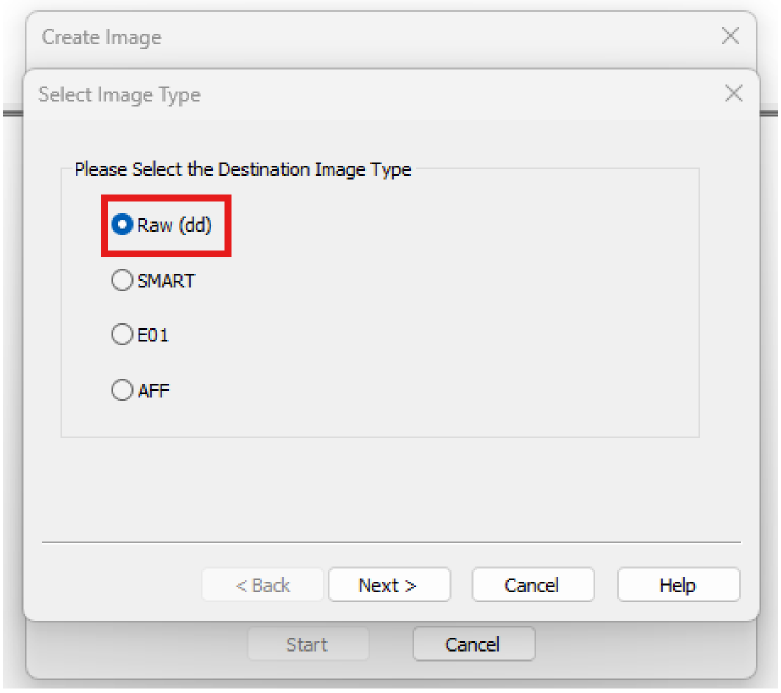


Figure 5: Selecting output Image type

Step 6:

Enter the details of the output image. No specific input is required in this step since the assignment didn't request any specific details about this part.

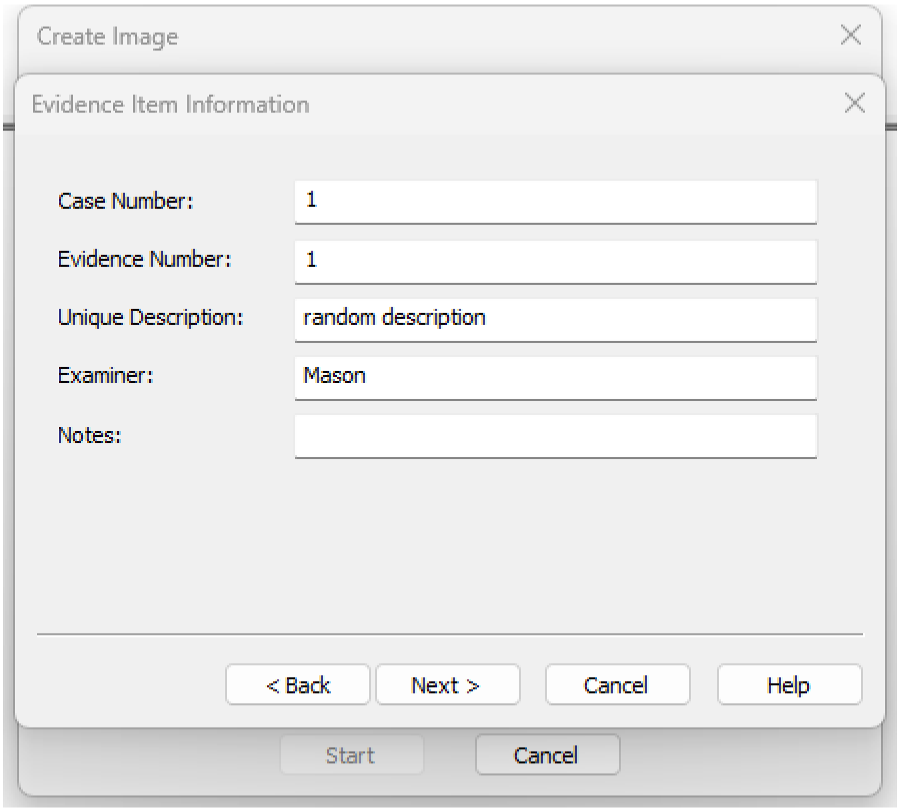


Figure 6: Configuring details of Image

Step 7:

For this step, put the image filename as requested within the assignment specifications. Additionally, change the fragment size from a default of 1500 to 0 (also requested by the assignment). Also, I've put the destination folder as my desktop, but this is only due to my preference and makes no impact on the result.

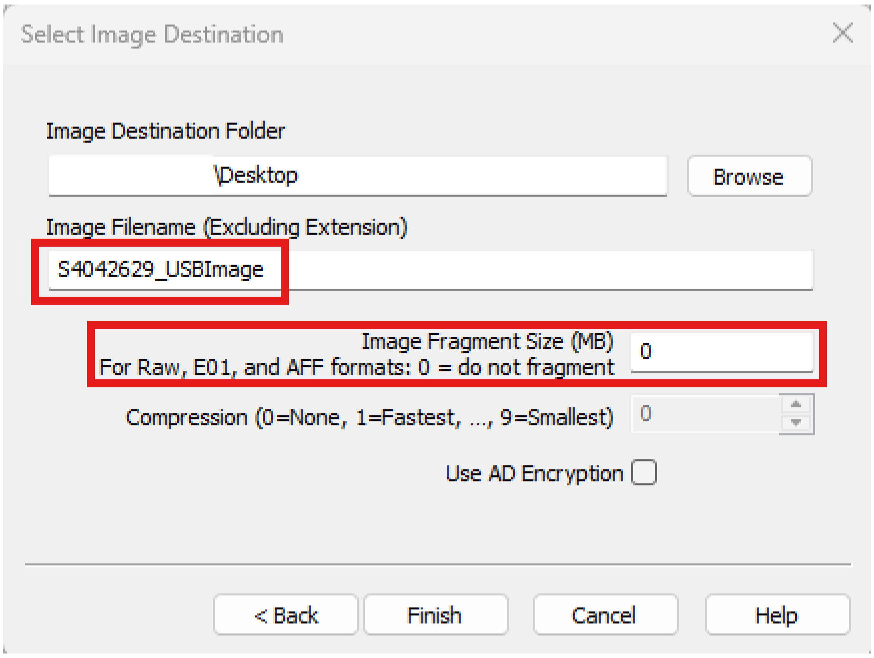


Figure 7: Configuring Image name, destination, and fragmentation

Step 8:

Recheck all the configuration of the disk image. Make sure "verify images after they are created" is ticked. Click "start" to start the process.

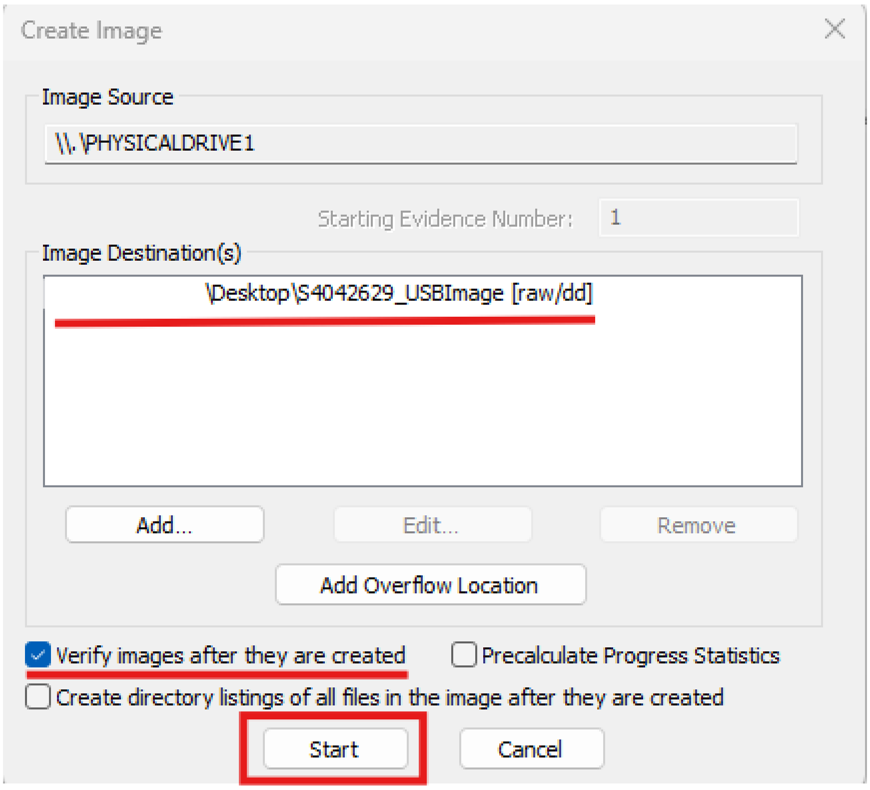


Figure 8: Starting Image creation

b. Disk Image information

The figure below showcases the information of the disk image, gathered from the .txt file.

```
Created By AccessData® FTK® Imager 4.7.1.2

Case Information:
Acquired using: ADI4.7.1.2
Case Number: 1
Evidence Number: 1
Unique description: random description
Examiner: Mason
Notes:

--------------------------------------------------------------

Information for C:[redacted]\Desktop\S4042629_USBImage:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 1,912
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 30,720,000
[Physical Drive Information]
 Drive Model: General UDisk USB Device
 Drive Serial Number: 0901090447033862834317
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 15000 MB
 Sector count:    30720000
[Computed Hashes]
 MD5 checksum:    5ca6503e059bd43306fd686b9fc33ce8
 SHA1 checksum:   f8a97c570222ff108163a33afa2fbcaf12b9e7cb

Image Information:
 Acquisition started:   Sat Aug  9 14:21:19 2025
 Acquisition finished:  Sat Aug  9 14:29:17 2025
 Segment list:
  C:[redacted]\Desktop\S4042629_USBImage.001

Image Verification Results:
 Verification started:  Sat Aug  9 14:29:18 2025
 Verification finished: Sat Aug  9 14:29:58 2025
 MD5 checksum:    5ca6503e059bd43306fd686b9fc33ce8 : verified
 SHA1 checksum:   f8a97c570222ff108163a33afa2fbcaf12b9e7cb : verified
```

Figure 9: Disk Image information

c. Questions

i. The creation process started on Aug 9 14:21:19 2025 and finished on Aug 9 14:29:17 2025. The process took roughly 8 minutes to complete.

ii.

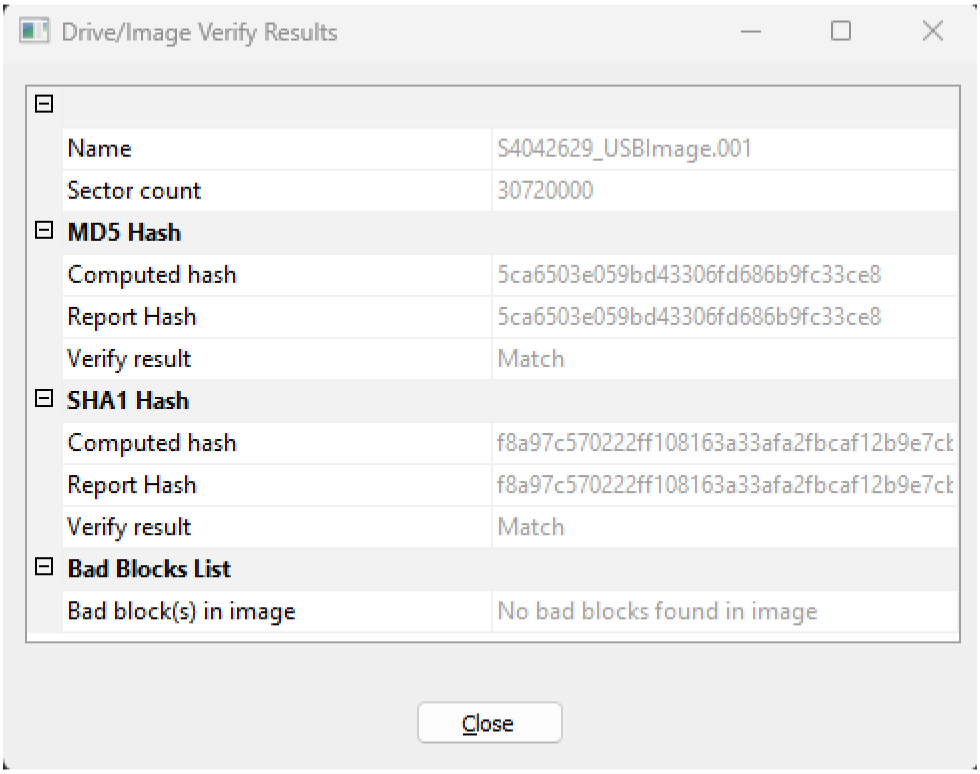


Figure 10: Image verification result

The checksums of the physical disk are:

- MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8
- SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb

iii. The checksums of the image are:

- MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8 : verified
- SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb : verified

The verification of both checksums was successful, as evidenced by the word “verified” next to each checksum of the image, along with the verify result for each checksum showing a match, as presented in Figure 10.

Q2. Disk Image Analysis

a. Case Creation

Step 1:

Select the first option named “New Case” to start a new case.

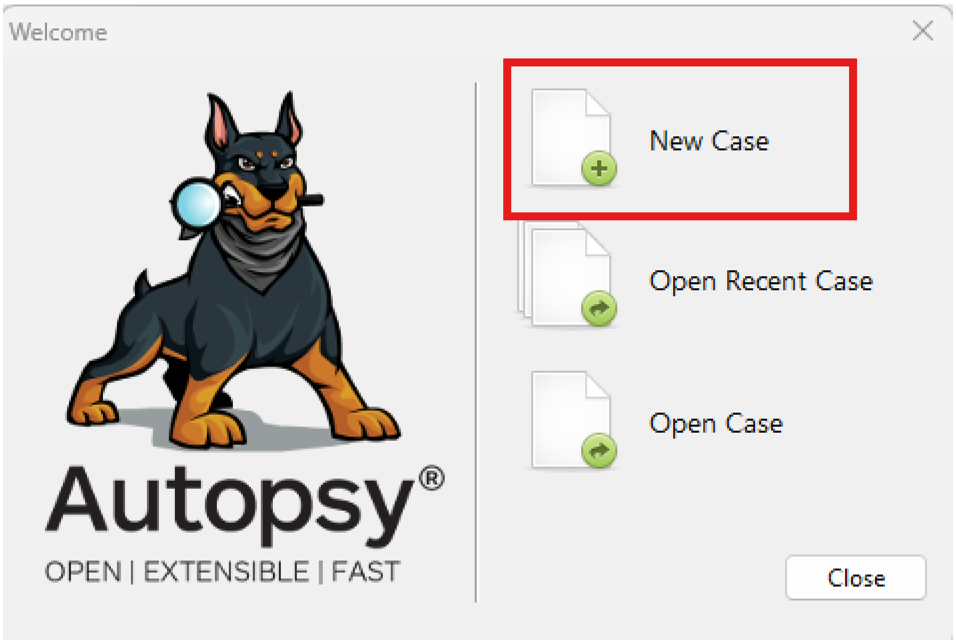


Figure 11: Creating new case

Step 2:

Enter the case name following the format described in the assignment details. Additionally, configure the base directory (where the case contents will be saved) if this is your first time creating a new case.

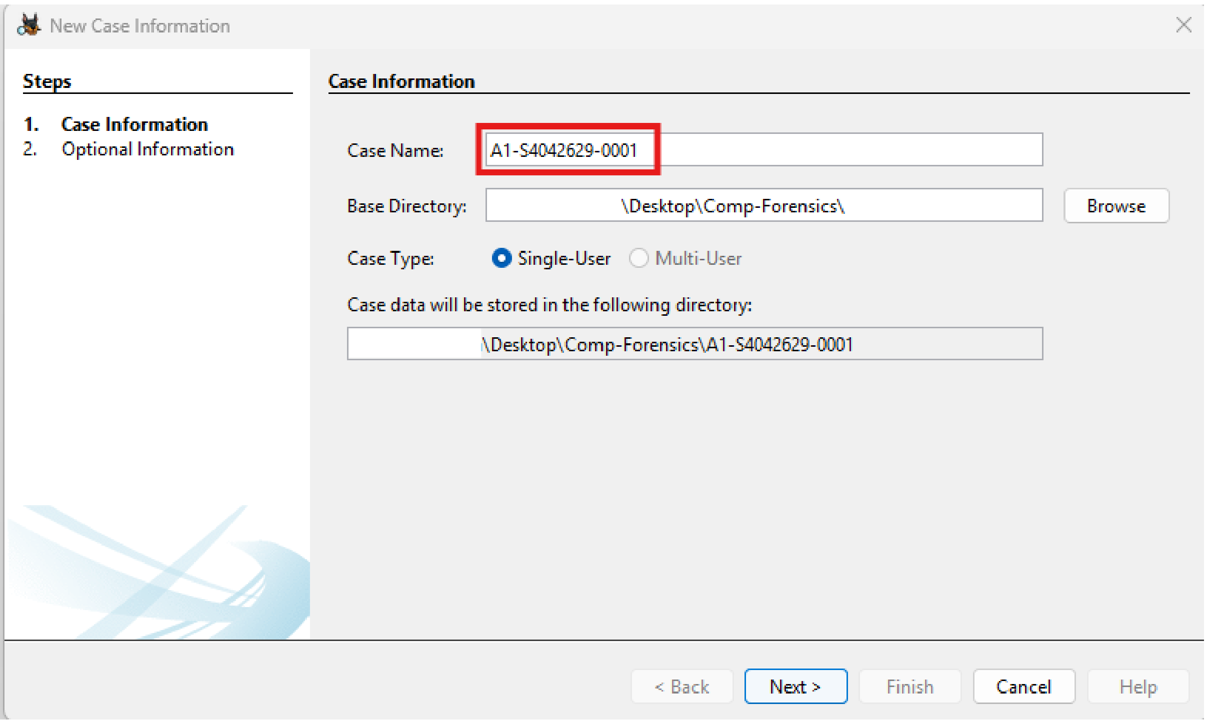


Figure 12: Configuring case name

Step 3:

Enter the requested optional information as shown in the figure above, then click "Finish" to proceed with the process.

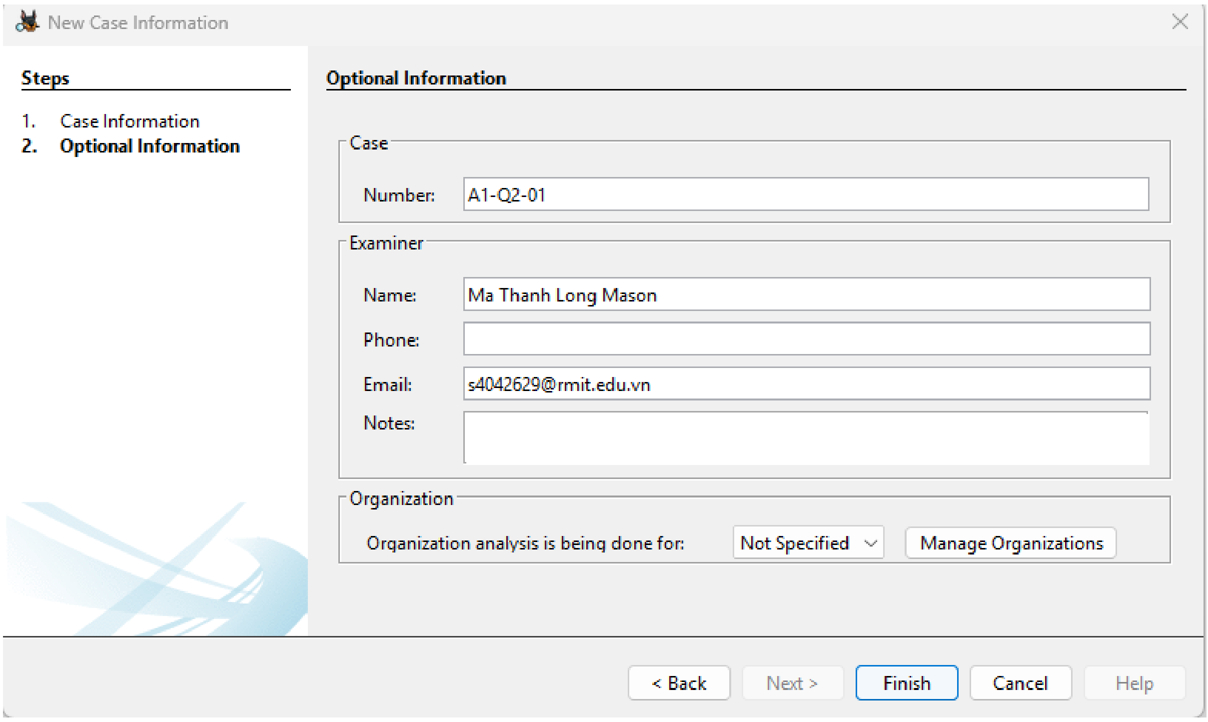


Figure 13: Configuring optional information

Step 4:

Select the second bullet point and enter in host name in the requested format.

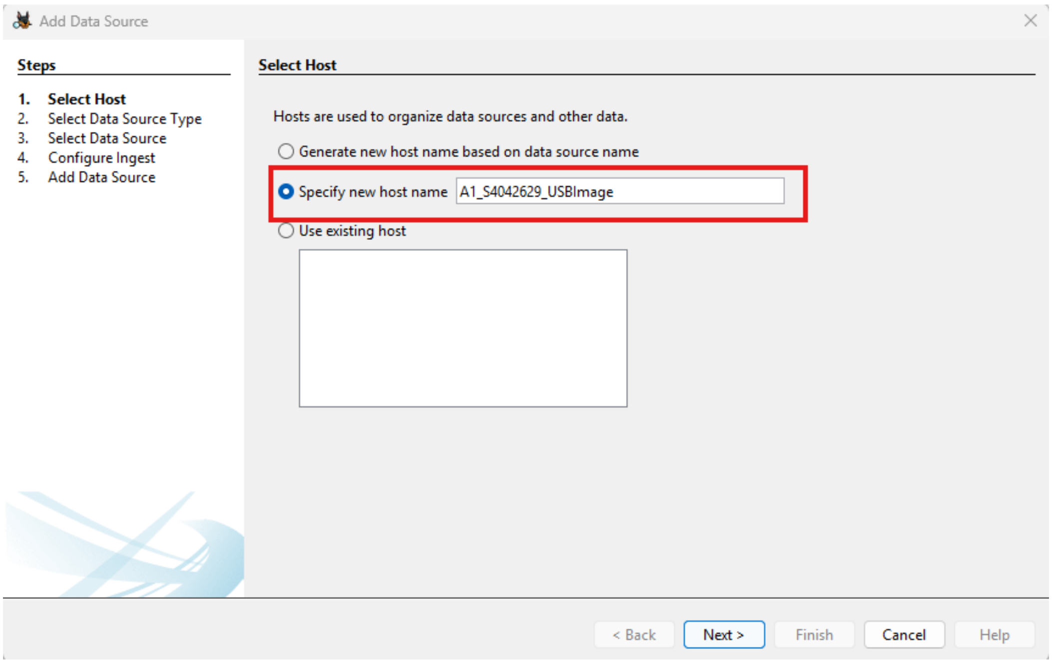


Figure 14: Configuring host name

Step 5:

Since we are trying to open and analyse a disk image (the one we created during Q1), we'll select the option "Disk Image or VM File" and click next to proceed.

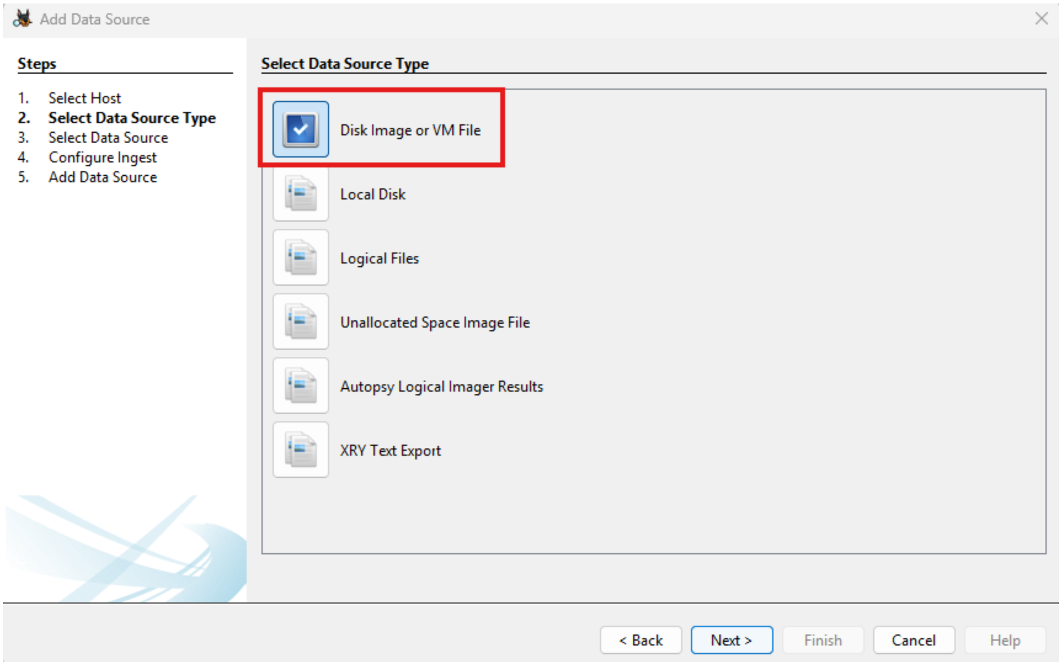


Figure 15: Selecting data source type

Step 6:

In this step, click on “Browse” and select the disk image created during Q1. Additionally, input the checksums into their appropriate text fields. Once finished, you can continue clicking on “Next” to proceed; leave everything in the next two pages as default and click on “Finish” once everything is done to conclude the case configuration.

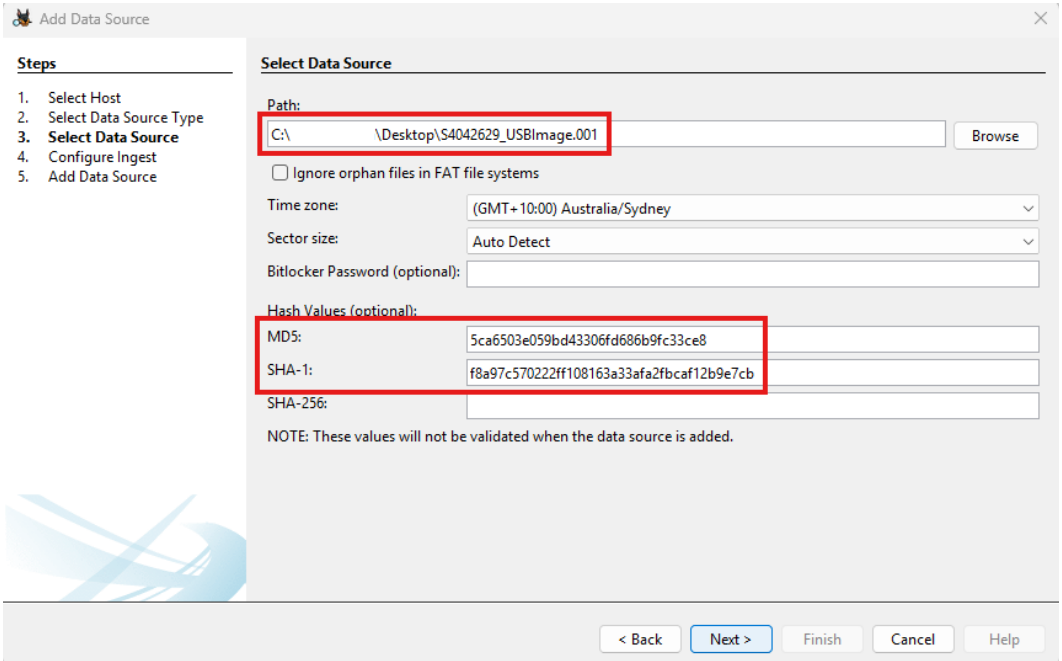


Figure 16: Entering information of data source

b. List of files

Table 1: List of files

File type	Amount
Images	3

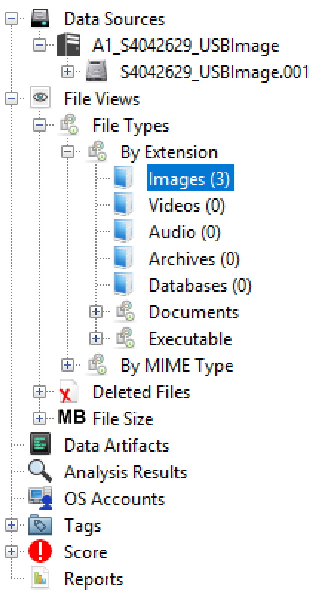


Figure 17: List of files

c. File extraction

Step 1:

Navigate the program by click on the various tabs, sections and locate the file you want to extract.

Figure 18: Navigating the Autopsy menu

Step 2:

Right click on the file you want to extract and select the “Extract File(s)” option to start.

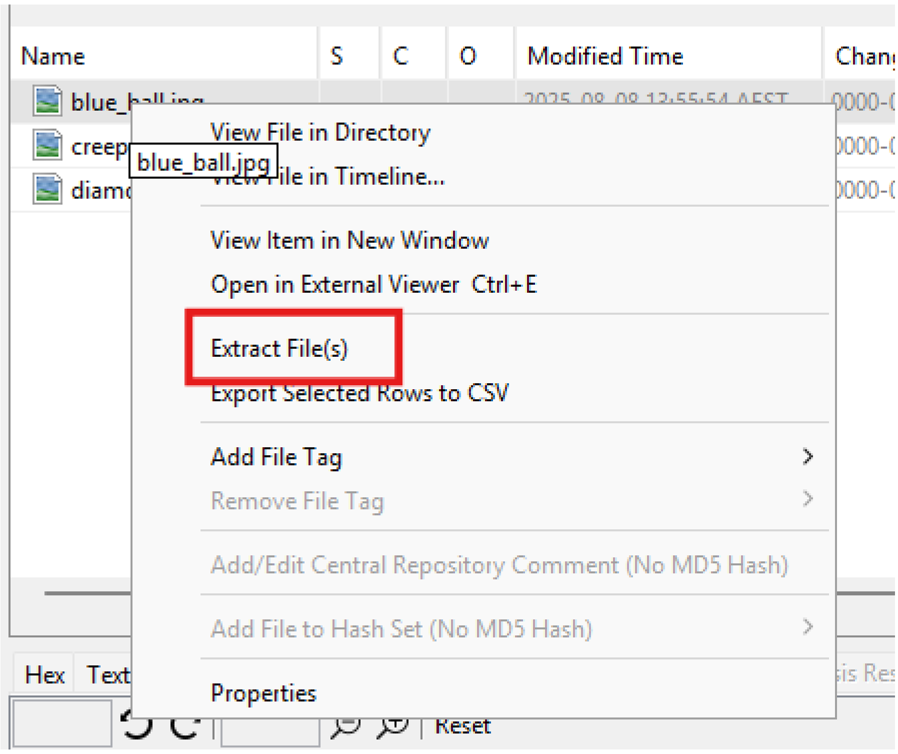


Figure 19: Extracting file

Step 3:

One you click on the "Extract File(s)" on the previous step, a window will appear for you to select the location of the extracted file. You can change the export location or leave it as default, which will save the extracted file to a sub-folder named "Export", located within the base directory of the case.

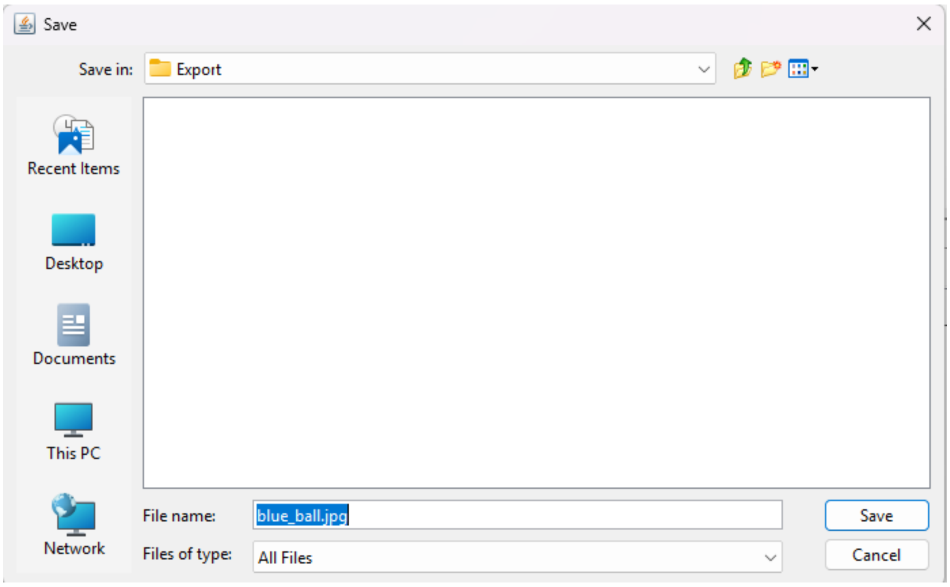


Figure 20: Saving extracted file

Q3. Understanding of Different Digital Forensics Tools

a. Static data acquisition tools

Tool	**dc3dd**	**Guymager**	**OSFClone**
OS	Linux & Windows	Linux	Linux & Windows
Pros	- Automatically detects bad sectors - Built-in hashing of input & output bytes - Integrates logging functionalities	- Can produce image files in raw, EWF, and AFF formats. - Multi-threaded - GUI-based program	- Bootable environment, independent of the installed operating system - Able to export in many formats such as: AFF, and EWF.
Cons	- CLI only, hard to use for beginners	- Linux only - Limited number of features beyond imaging	- May not be forensically sound when imaging drives with ext2/3/4 filesystems - No integrated analysis tools.

Table 2: Static data acquisition tools

b. Memory acquisition tools

Tool	**DumpIt**	**LiME**	**WinPmem**
OS	Windows	Linux & Andriod	Linux & Windows
Pros	- Portable, full physical memory capture - Minimal user interaction needed	- Acquires memory to file or over network - Supports ARM which is useful for mobile forensics - Open-source	- Supports multiple format - Integrated with Rekall for direct analysis - Open-source
Cons	- Windows-only - Always captures full RAM, which may not always be desired	- Requires loading a kernel module - Difficult to use	- Mostly CLI-based, not beginner-friendly - Has MacOS support but severely limited

Table 3: Memory acquisition tools

Q4. Disk Image Artifact Analysis

a. File Listing and carving

Step 1:

Create a new case to start analysing the disk image. You can follow Figure 21 and Figure 22 to see what to put into the various input fields, but these are purely optional and doesn't impact the disk image analysis process.

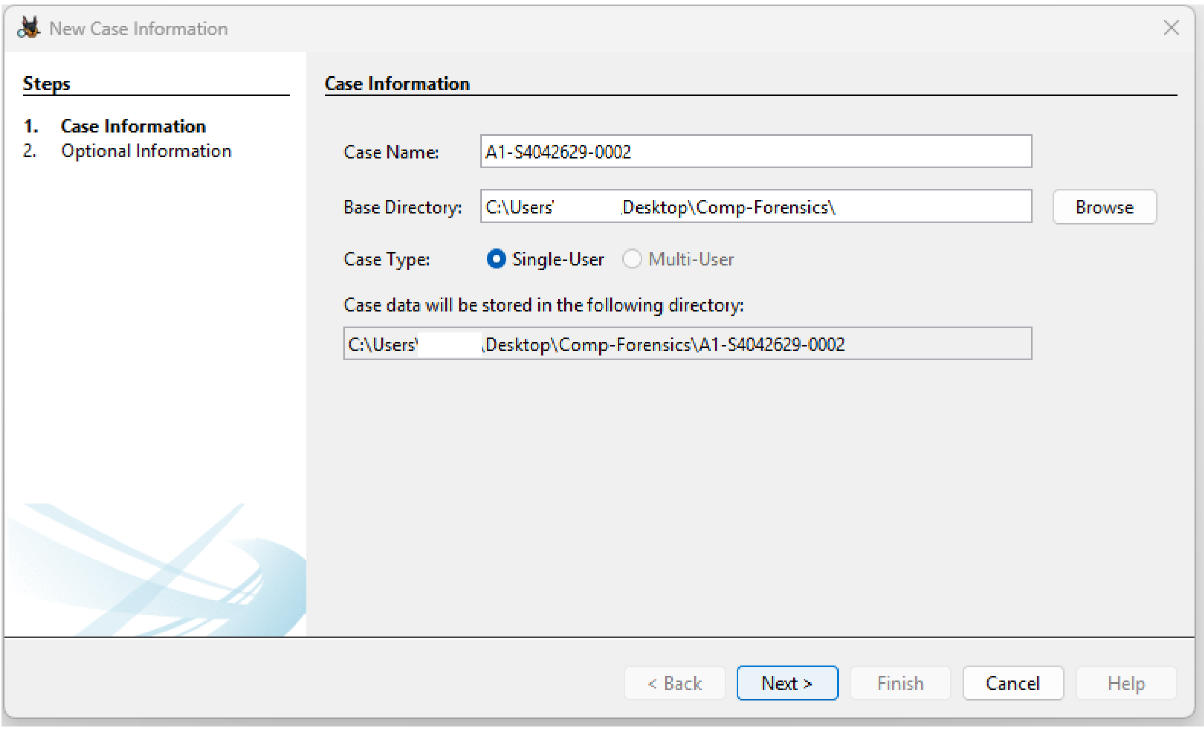


Figure 21: Creating a new case

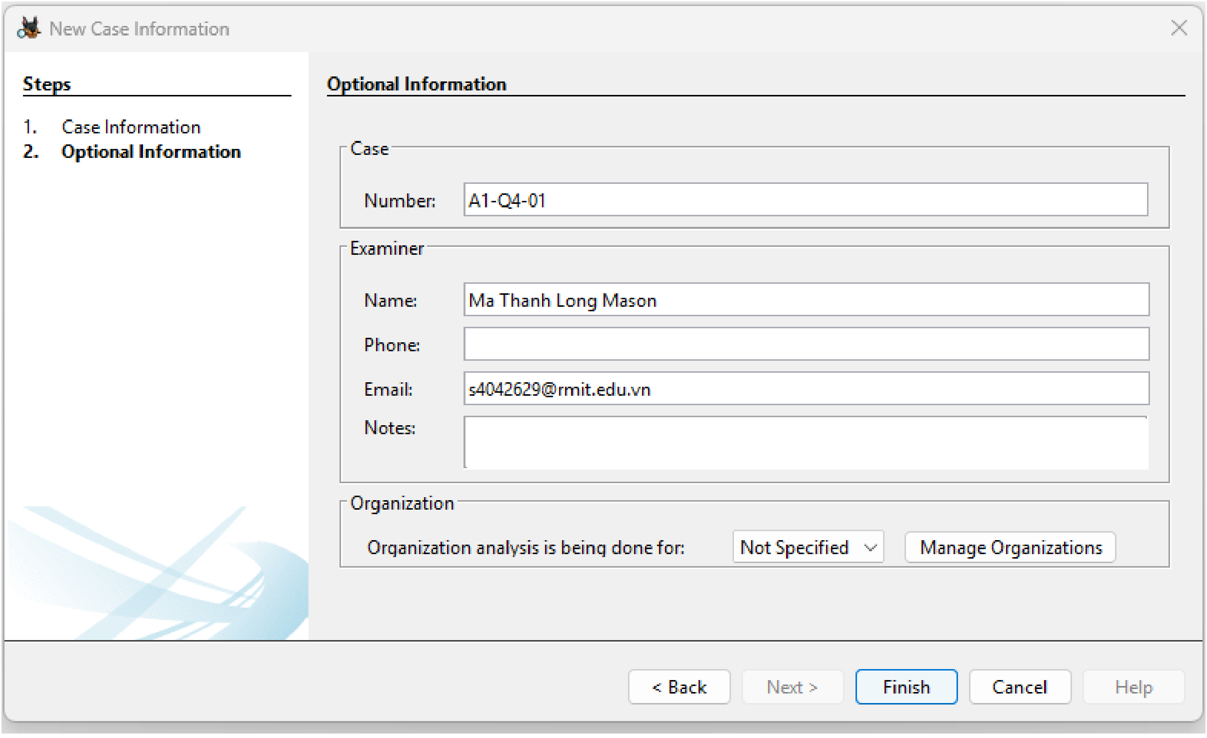


Figure 22: Entering optional information

Step 2:

Figures 23, 24, and 25 showcase the process of configuring the host and data source of the case. Follow the selected options in figures 23 and 24, which should be already selected by default, and for figure 25, click the browse button and select the disk image file.

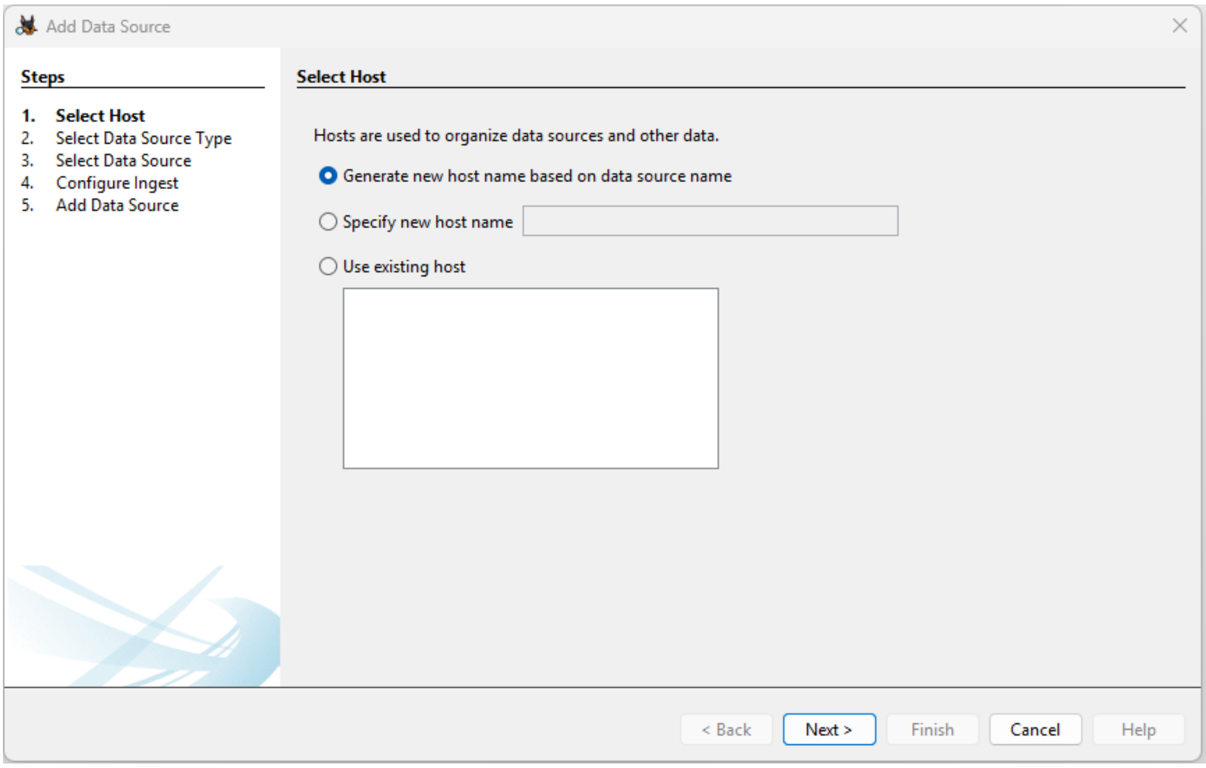


Figure 23: Selecting host

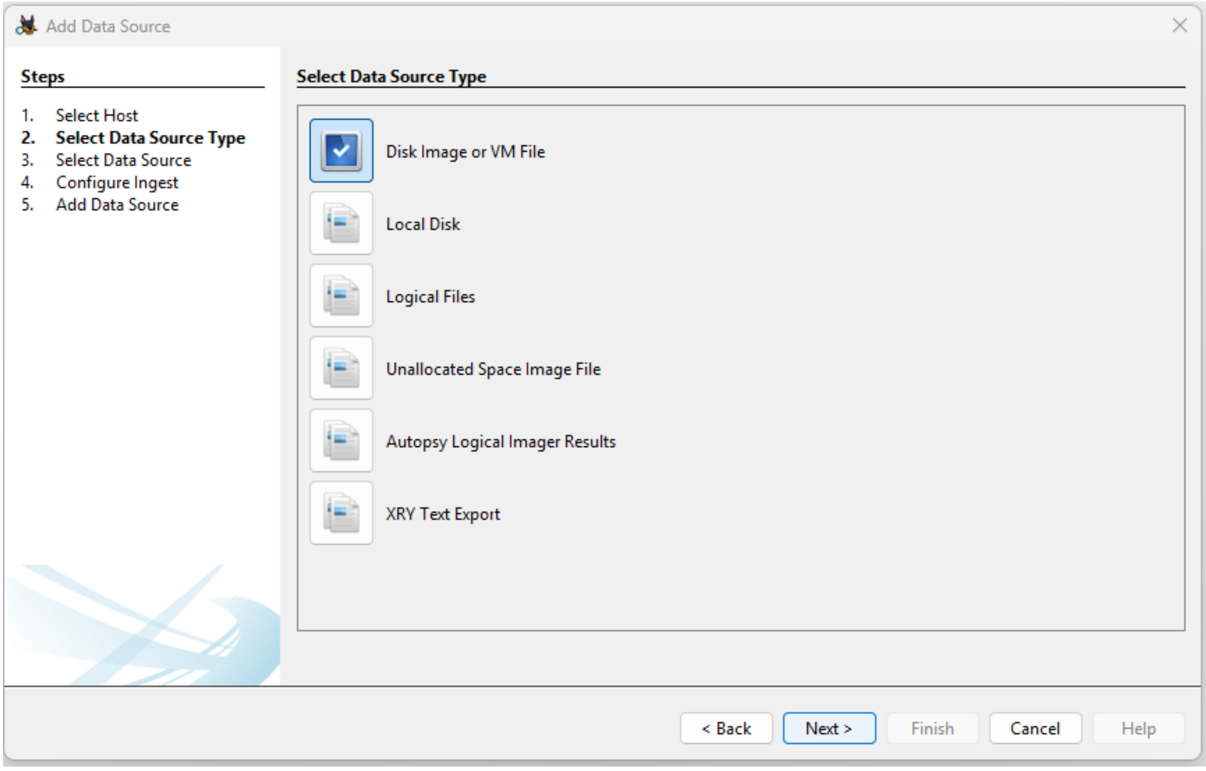


Figure 24: Selecting data source type

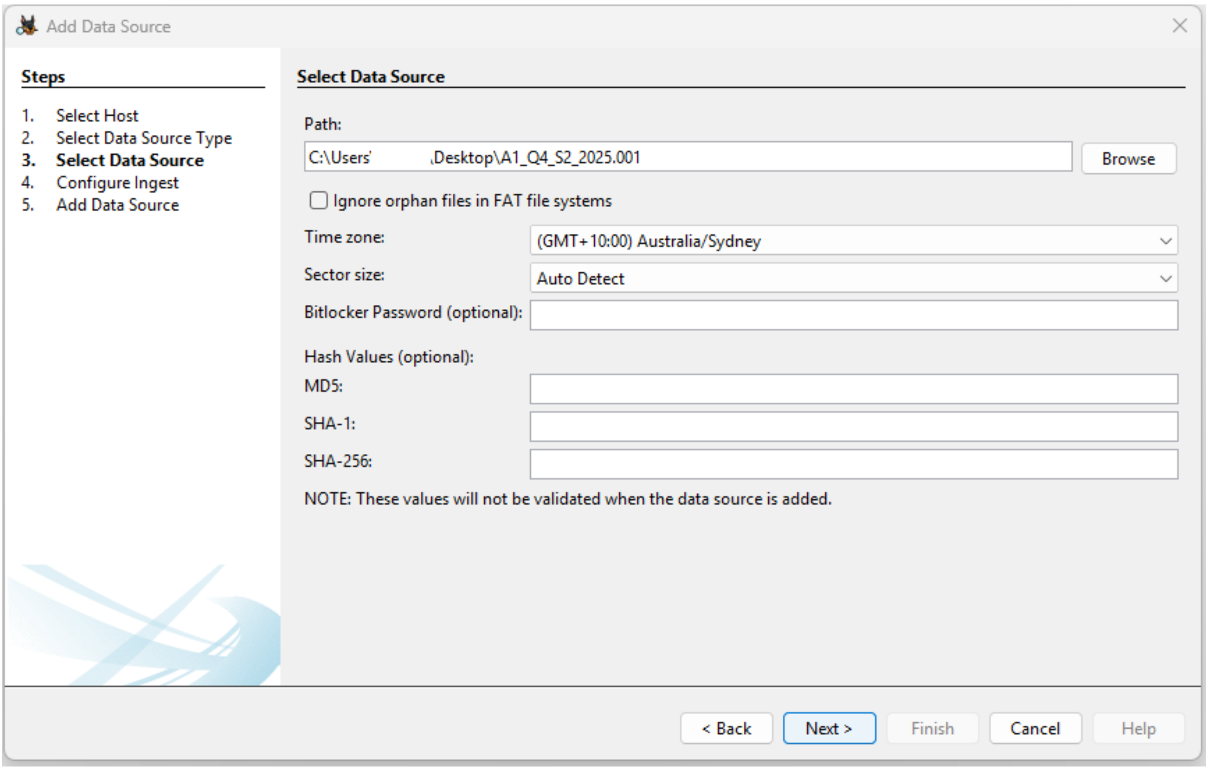


Figure 25: Selecting data source

Step 3:

Leave everything in this step as default, which should have all the modules selected except for "Cyber Triage Malware Scanner" and "Plaso". Once you are done with this step, click "next" and on the next page click "Finish" to complete the case creation.

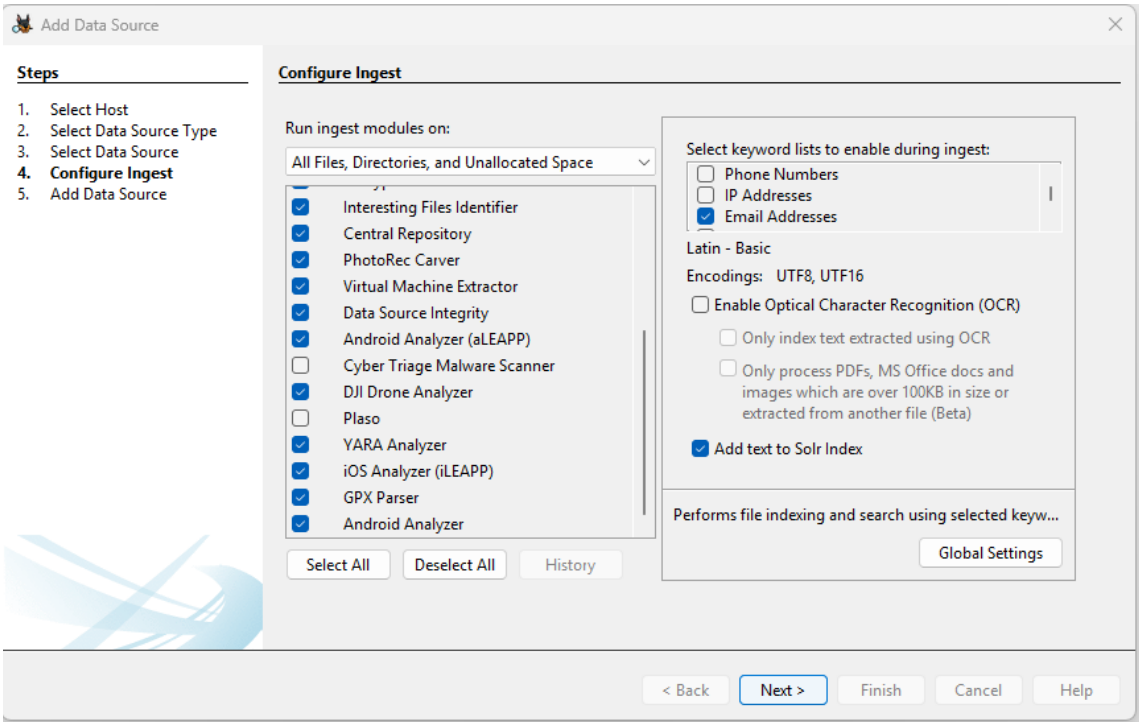


Figure 26: Configuring ingestion

Step 4

Once you have reached this step, you should now see the file contents of the disk image. However, as shown in figure 27, some tabs like the "By MIME Type" tab is missing its data, which hints at the ingestion being incomplete.

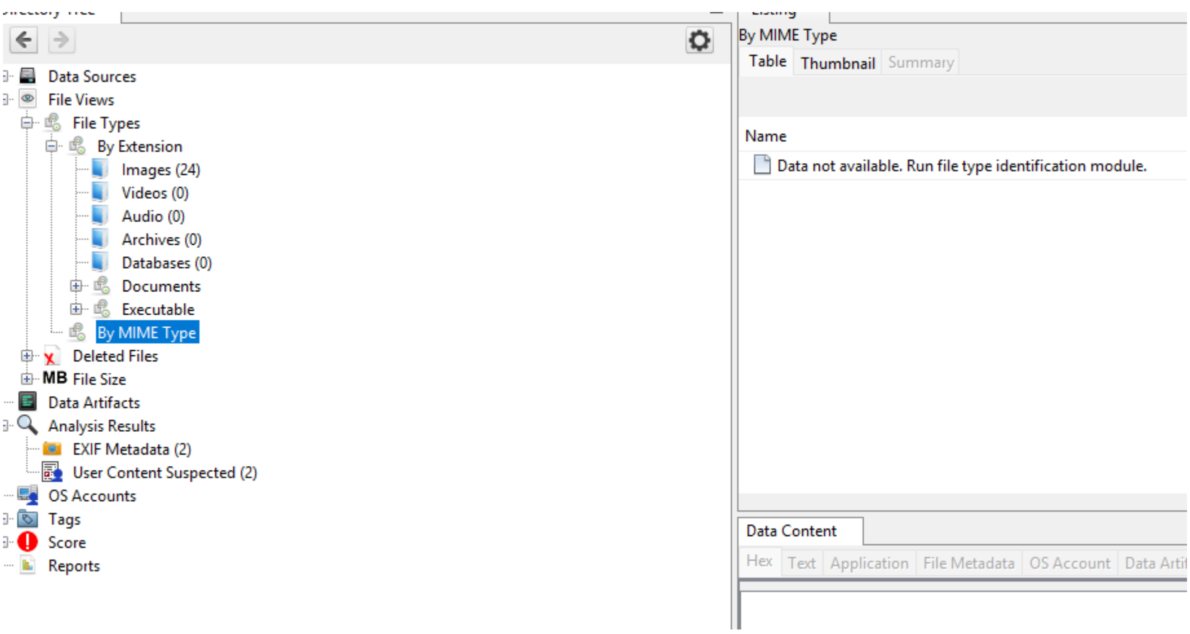


Figure 27: Disk image contents first look

Step 5:

We can try to fix this by navigating to the "tools" tab at the top of screen, navigating to "Run Ingest modules" and selecting the disk image file, as shown in figure 28.

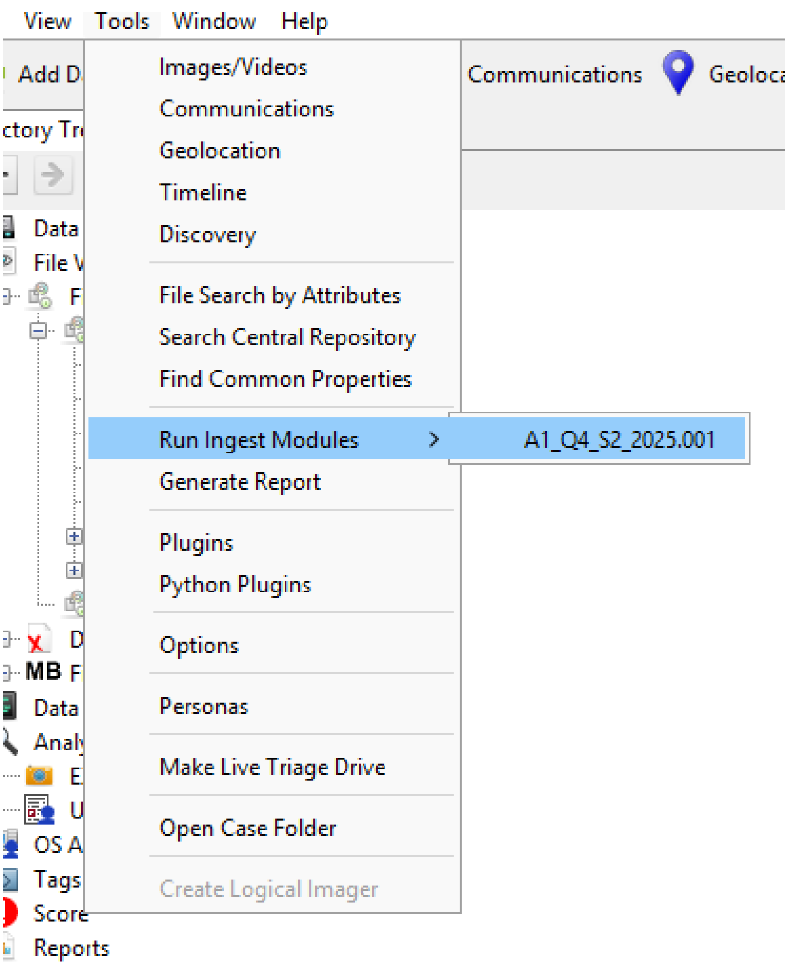


Figure 28: Running ingestion modules

Step 6:

Leave everything in the “Configure Ingest Modules” tab as default, as shown below, and click “Finish” to proceed with the ingestion.

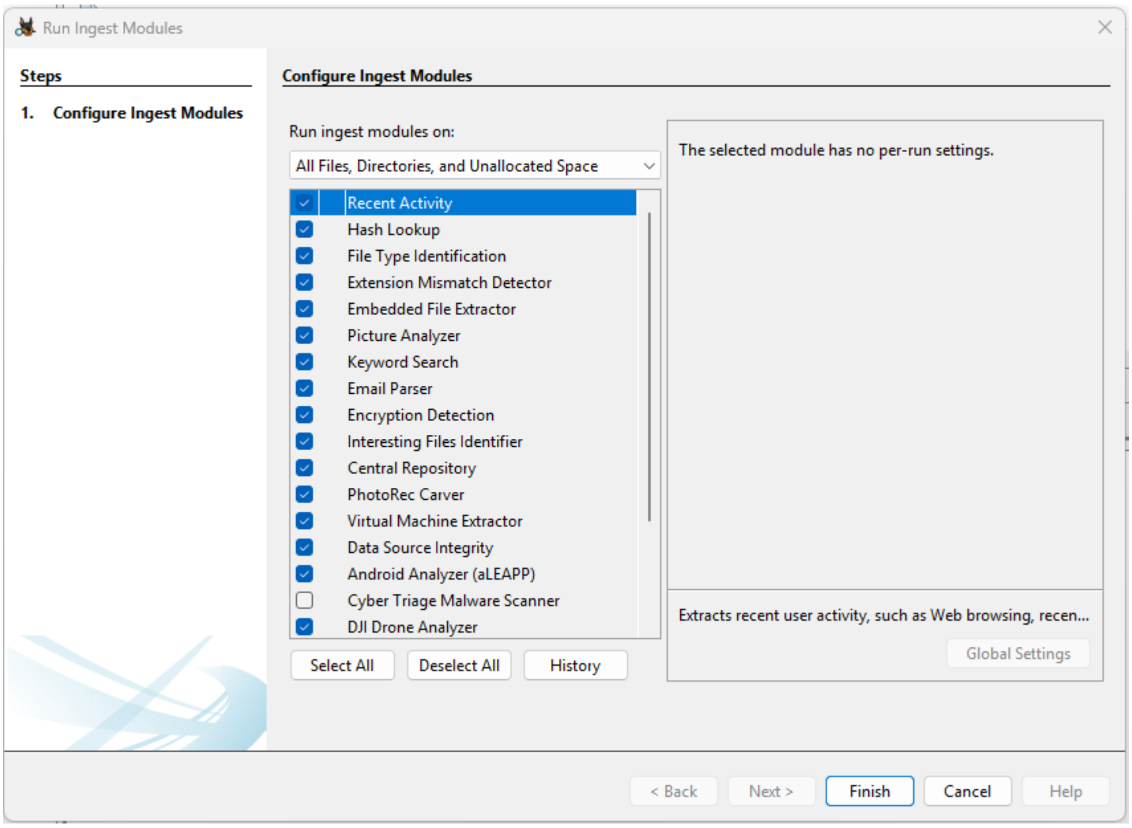


Figure 29: Configuring ingestion modules

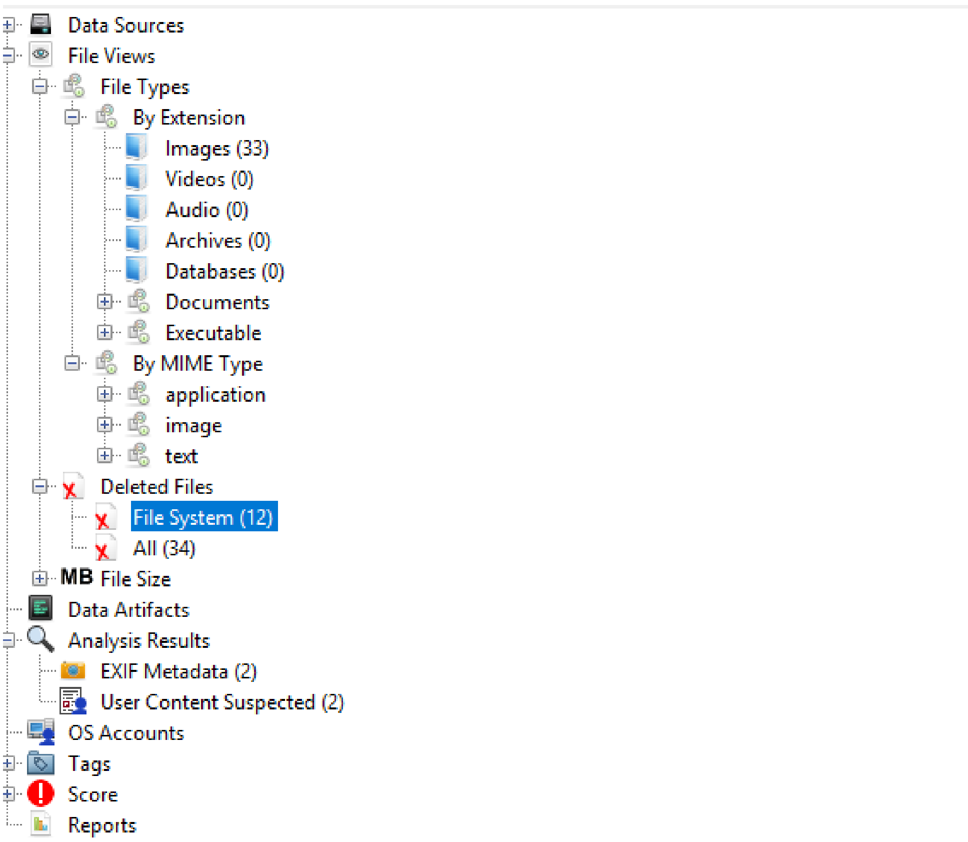


Figure 30: Complete ingestion of disk image

List of files by extension

Images

- animal5.jpg
- animal2.jpg
- animal6.jpg
- hashAlgo.jpg
- myfav.jpeg
- myfav.jpg
- f0000016.jpg
- f0000032.jpg
- f0000016.jpg
- f0000032.jpg
- animal1.png
- caesarcipher.png
- interesting1.png
- ref1.png
- rotCipher.png
- secretcode.png
- stag1.png
- stago1.png
- tools.png
- f0000056.png
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png
- f0000056.png
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png

Plain text

- AES.txt
- log1.txt

- log3.txt
- f0000392.txt
- f0000392.txt

List of files by MIME type

application/octet-stream

- $AttrDef
- $BadClus
- $Bitmap
- $Boot
- $ObjId
- $Quota
- $Reparse
- $Repair
- $Repair:$Config
- $Tops:$T
- $TxfLog.blf
- $TxfLogContainer00000000000000000001
- $TxfLogContainer00000000000000000002
- $LogFile
- $MFT
- $MFTMirr
- $Volume
- Bankdetails.gpg
- IndexerVolumeGuid
- WPSettings.dat
- f0000000.DS_Store
- f0000000.DS_Store

image/vnd.microsoft.icon

- $UpCase

image/vnd.zbrush.pcx

- $Tops

List of deleted files

- .DS_Store
- AES (1).txt
- animal3.jpg
- animal4.jpg

- interesting.txt
- log2.txt
- ref2.png
- ref5.png
- saf1.png
- secretPlan.png
- stago_howitworks.png
- useful.png
- f0000000.DS_Store
- f0000016.jpg
- f0000032.jpg
- f0000056.png
- f0000392.txt
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png
- f0000000.DS_Store
- f0000016.jpg
- f0000032.jpg
- f0000056.png
- f0000392.txt
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png

b. Finding Suspect’s Hideout Location

Step 1:

Preform a keyword search to identify the suspicious files.

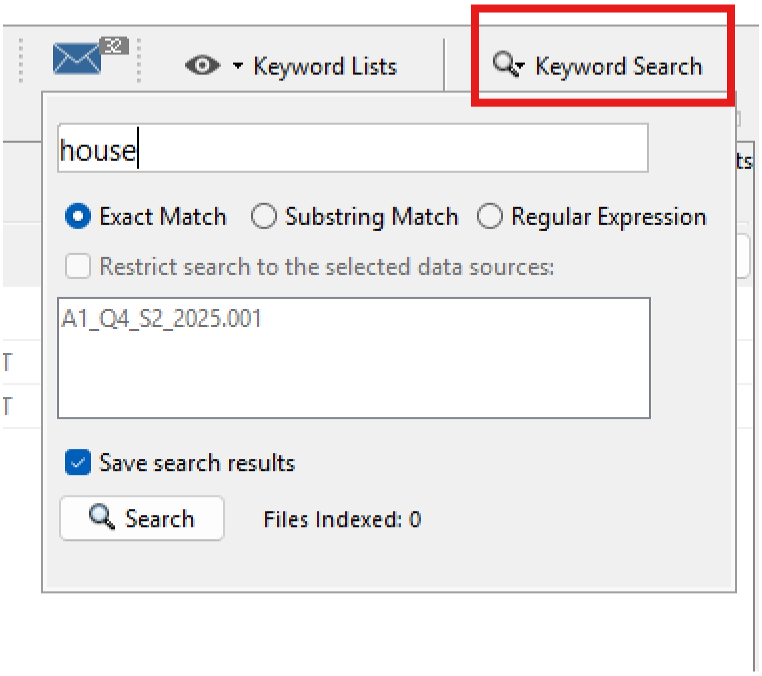
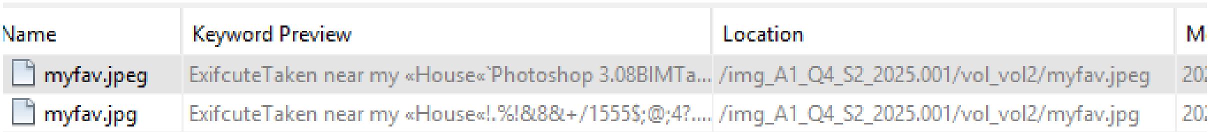


Figure 31: Keyword Search



Name	Keyword Preview	Location	M
myfav.jpeg	ExifcuteTaken near my «House«`Photoshop 3.08BIMTa...	/img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpeg	202
myfav.jpg	ExifcuteTaken near my «House«!.%!&8&+/1555$;@;4?....	/img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpg	202

Figure 32: Suspicious files

Step 2:

Navigate to the suspicious files and select the "Analysis Results" tab. In said tab, the latitude and longitude of where the picture was taken should be displayed if available.

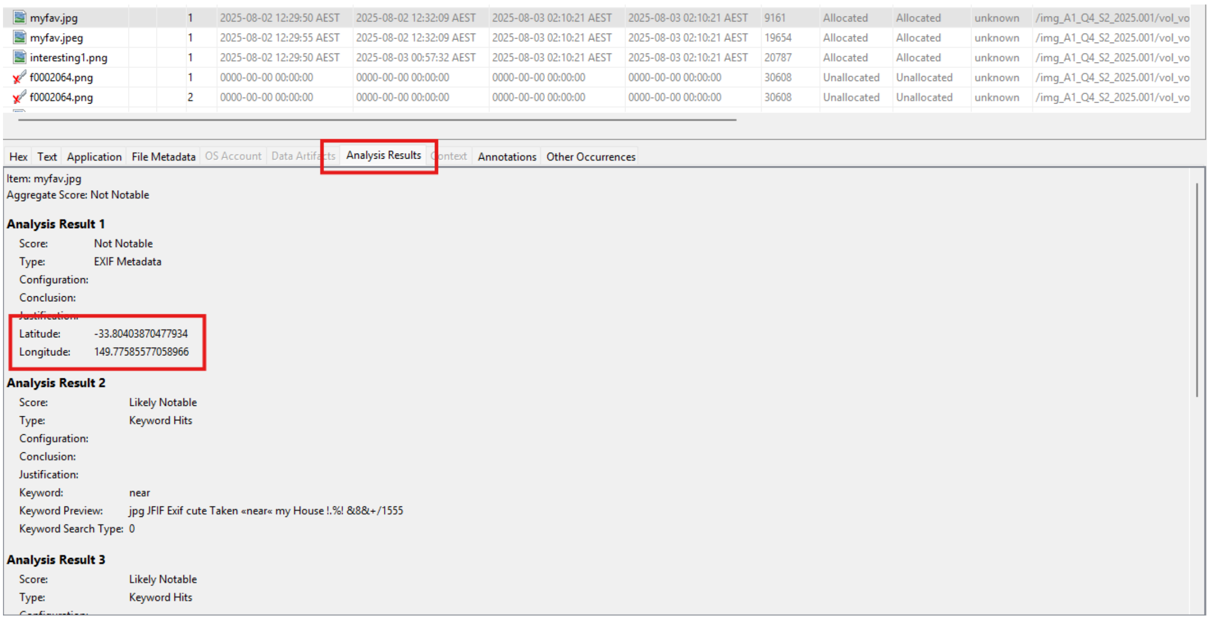


Figure 33: Coordinates in image

Result

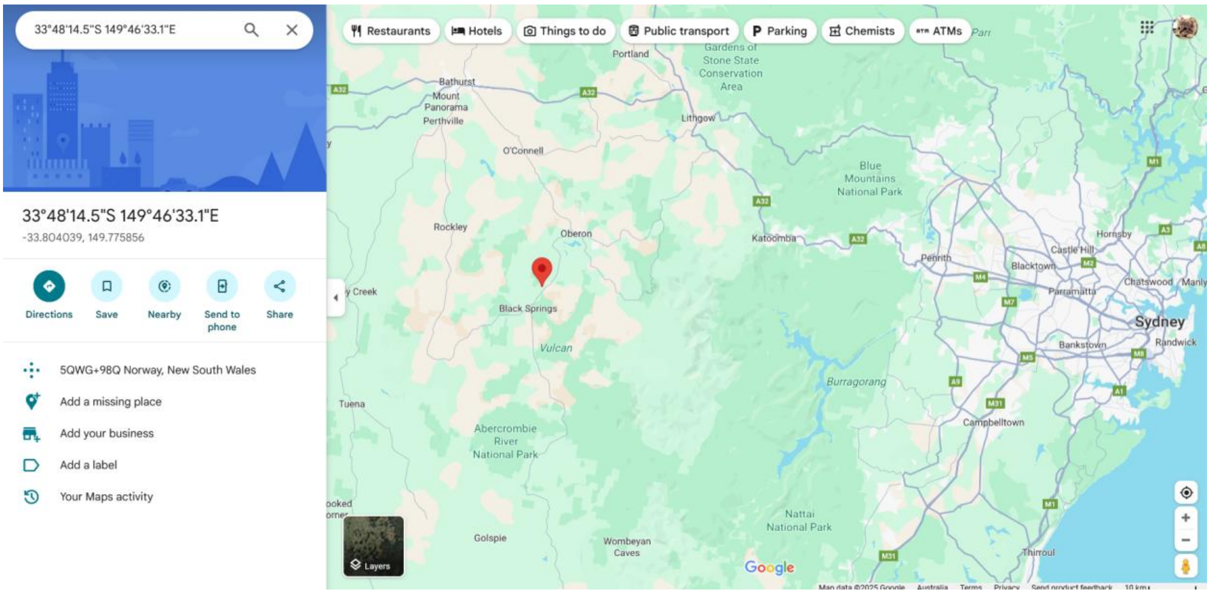


Figure 34: Extracted coordinates in google maps

c. Decryption of Bank Login Details

Step 1:

We first locate the suspicious file by browsing through all the different files on the disk image and isolating the most promising one.

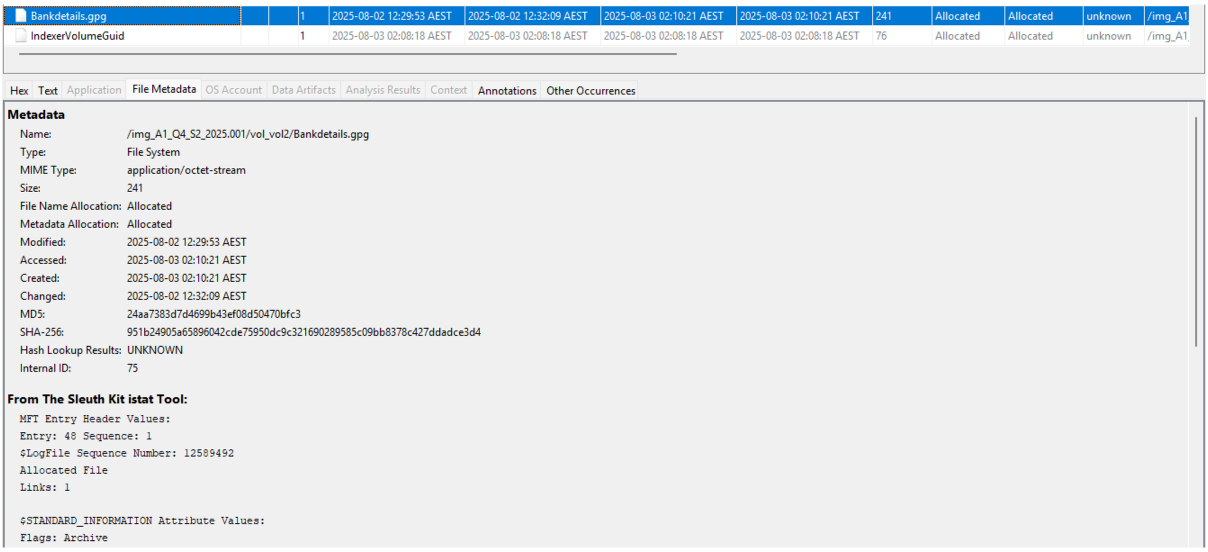


Figure 35: Potential bank details

Step 2:

Once we have located the suspicious file, we will browse through the disk image again to find clues on how to decrypt the file. The image shown in the figure below shows a command line that matches the tail ".gpg" of the suspicious file. By analysing the command, we can deduce that the file was possibly encrypted using symmetric encryption thus requires a passphrase to decrypt. Furthermore, we can see that AES256 was the algorithm used to encrypt the file.

Figure 36: Encryption command

Step 3:

A file named "AES", located within the plain text section could potentially be related to the command shown in figure 36. We can try brute forcing the answer by putting the extracted text into different cipher-solving tools to decrypt the text. The text was encrypted using the ROT cipher as shown in the two figures below.

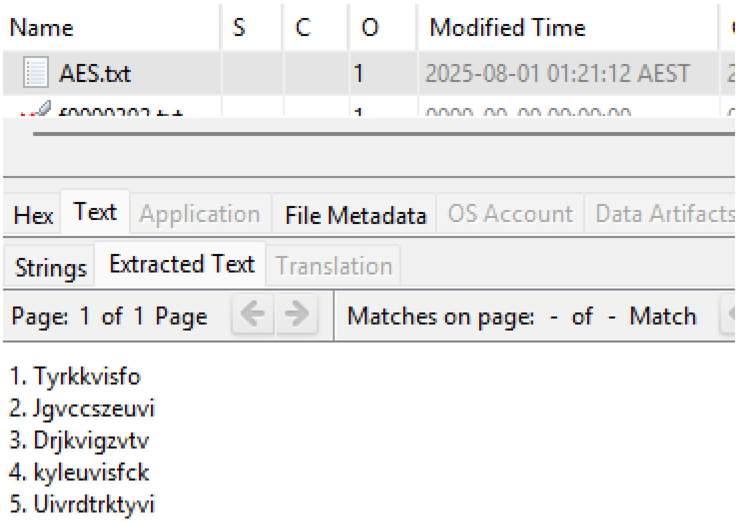


Figure 37: Encrypted passphrase

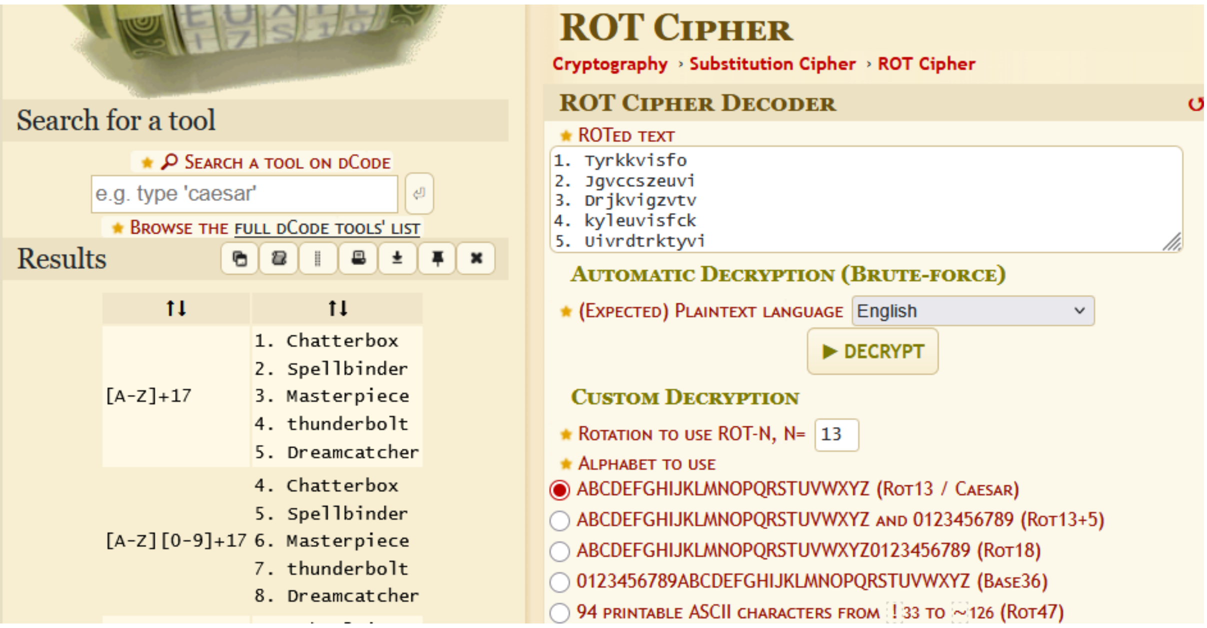


Figure 38: Decrypted passphrase

Step 4:

Another image file shows the command to decrypt the bankdetails file. We can proceed by going to KaliLinux and running the command, substituting the "sample.txt.gpg" with "Bankdetails.gpg" as shown in the two figures below. Once we run the command, the system will ask us for a passphrase to decrypt the file. We can now try to brute force the passphrase by trying all the 5 phrases we got from decrypting the "AES" text file.

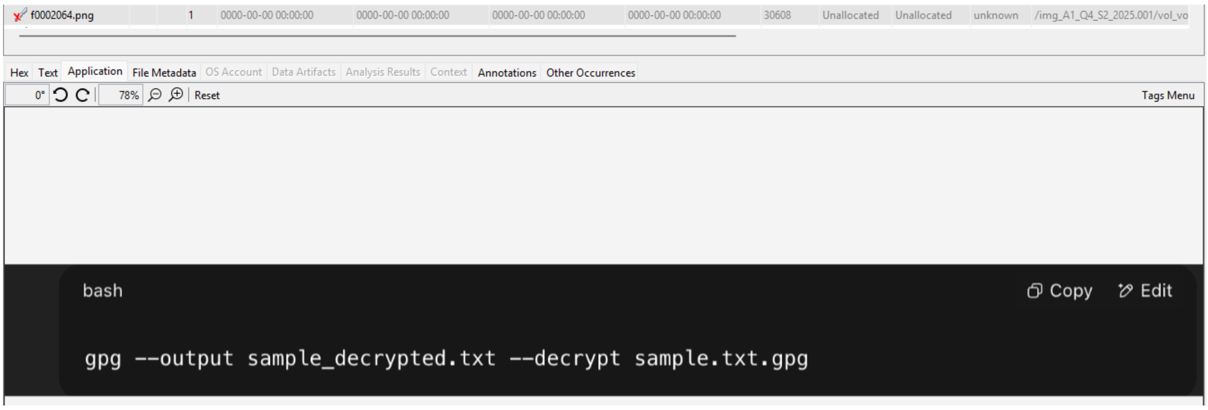


Figure 39: Gpg decrypt command

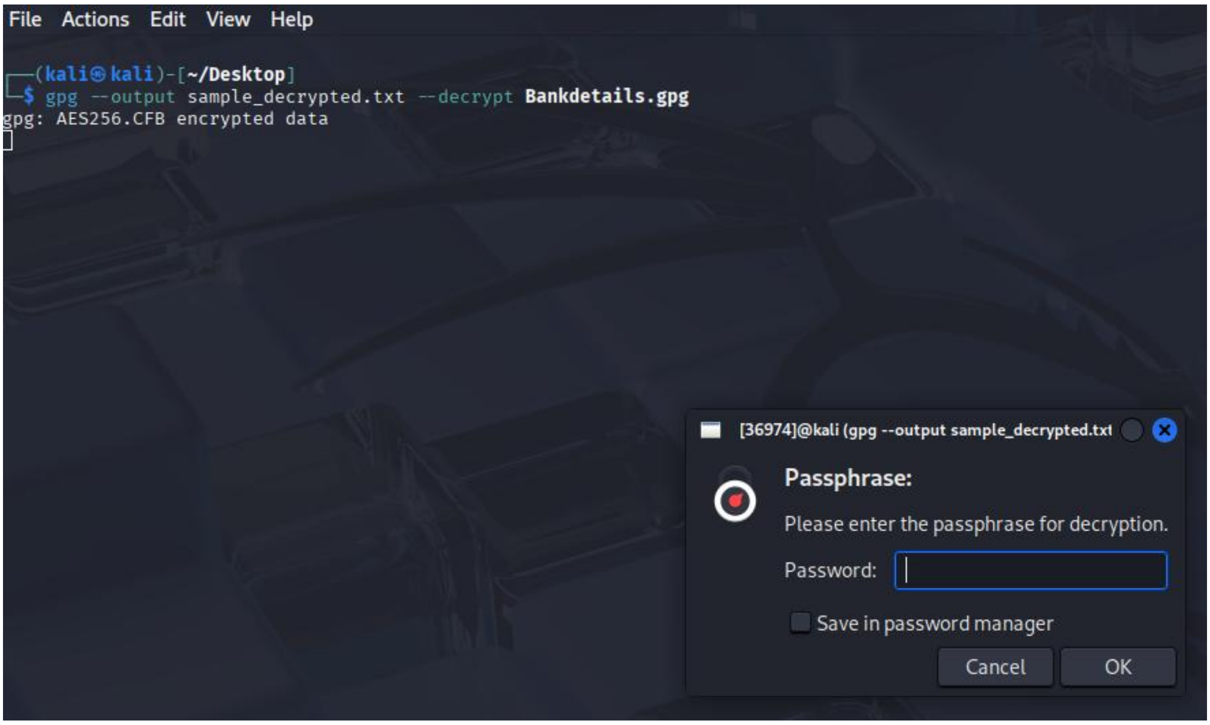


Figure 40: Prompted for passphrase

Result

The correct passphrase was "thunderbolt". The contents of the file is shown in the figure below.

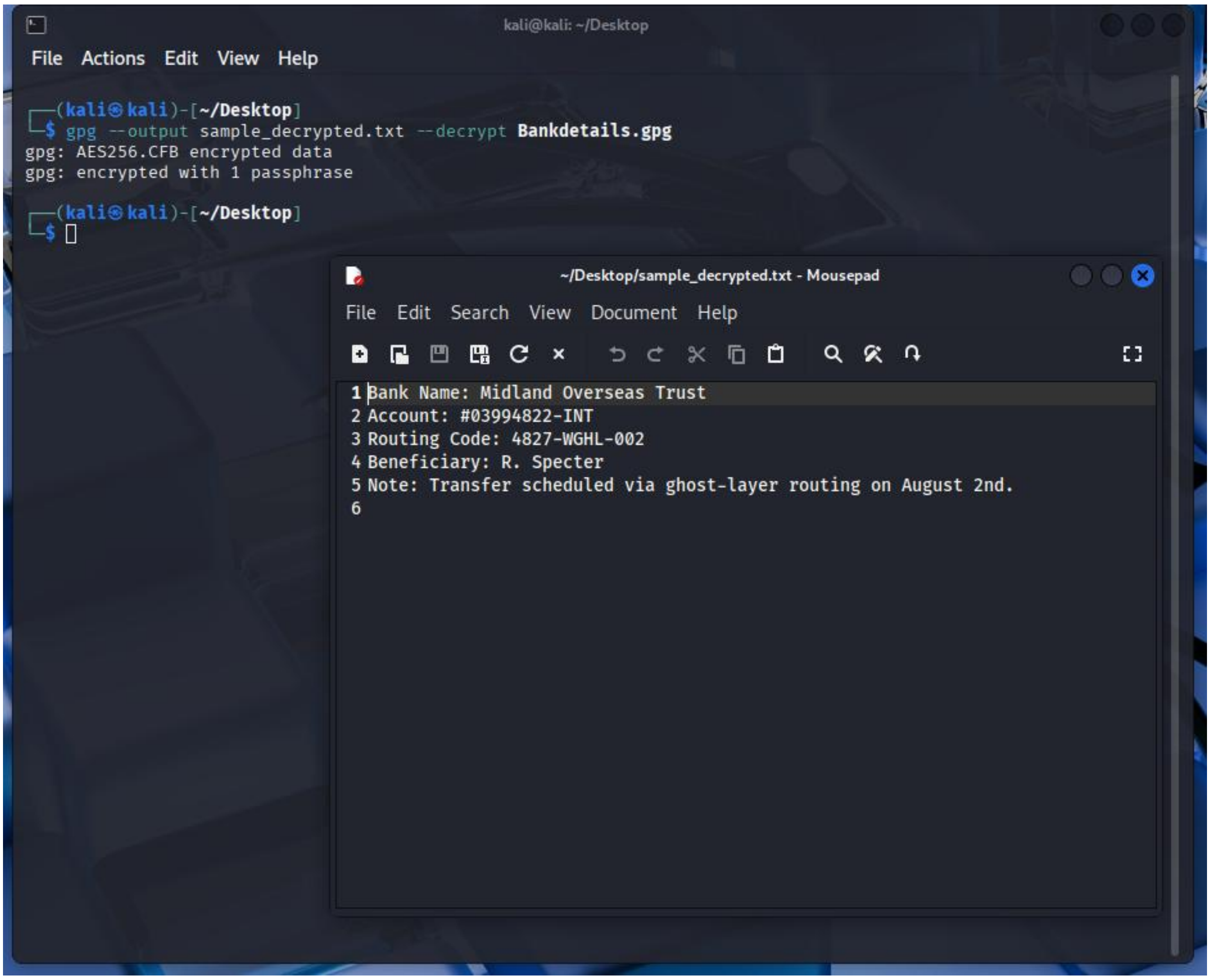


Figure 41: Decrypted bank details

d. Recovering the Vault PIN and Drawer Password

Step 1:

By browsing through all the files present in the disk image, we can find a suspicious .txt file which displays an image when we click on it.

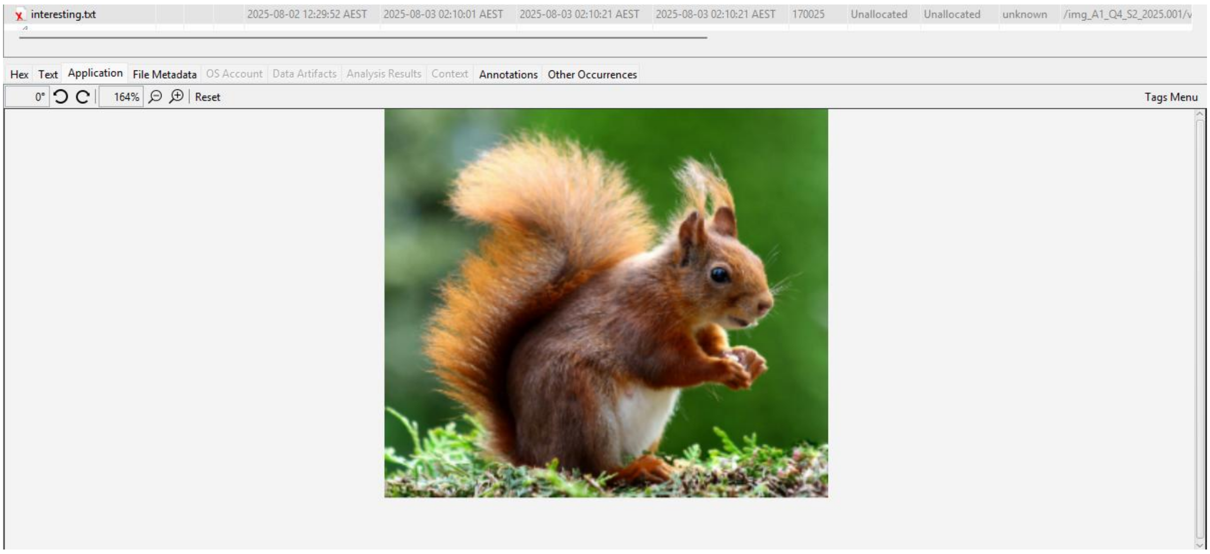


Figure 42: Suspicious file

If we examine the MIME type in metadata of the file, we can see that the file is in fact a .png file instead of a .txt file.

Metadata

Name:	/img_A1_Q4_S2_2025.001/vol_vol2/interesting.txt
Type:	File System
MIME Type:	image/png
Size:	170025

Figure 43: Mismatching MIME Type

Step 2:

After exporting the suspicious file and running it through a steganography decrypting tool, we can see the hidden picture.

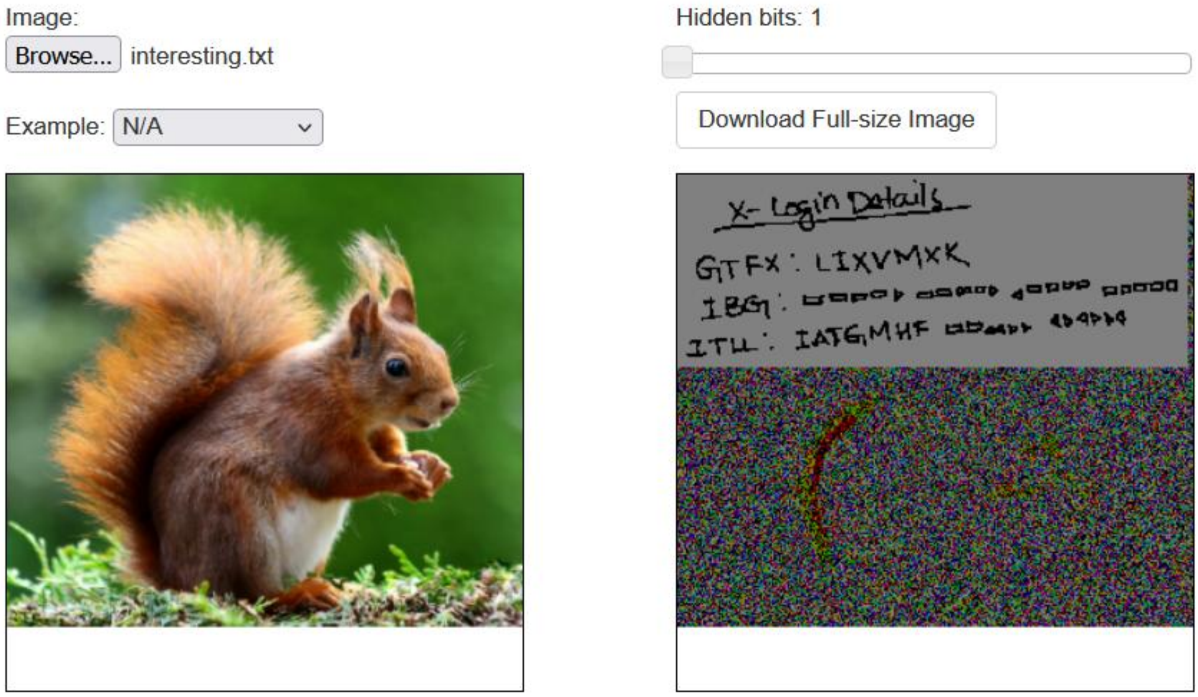


Figure 44: Decrypting image

Step 3:

We’ll first decrypt the squares and triangles by using morse code. By mapping the hidden message onto the table provided in the figure below, we can make out the hidden message on the second and third line (excluding the login details line) to be “9910” and “75” respectively.

Figure 45: Morse code table

Step 4:

We'll now decrypt the remaining message using the Caesar cipher to get the final decrypted message.

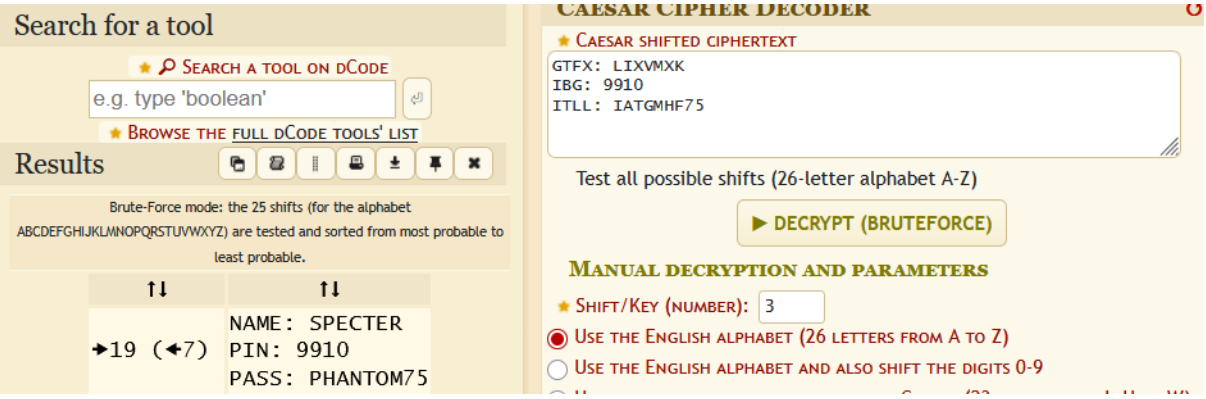


Figure 46: Decrypting hidden text

Result

NAME: SPECTER

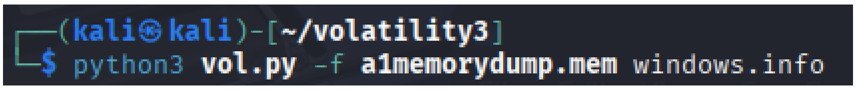
PIN: 9910

PASSWORD: PHANTOM75

Q5. Memory Data Analysis from Computer Memory

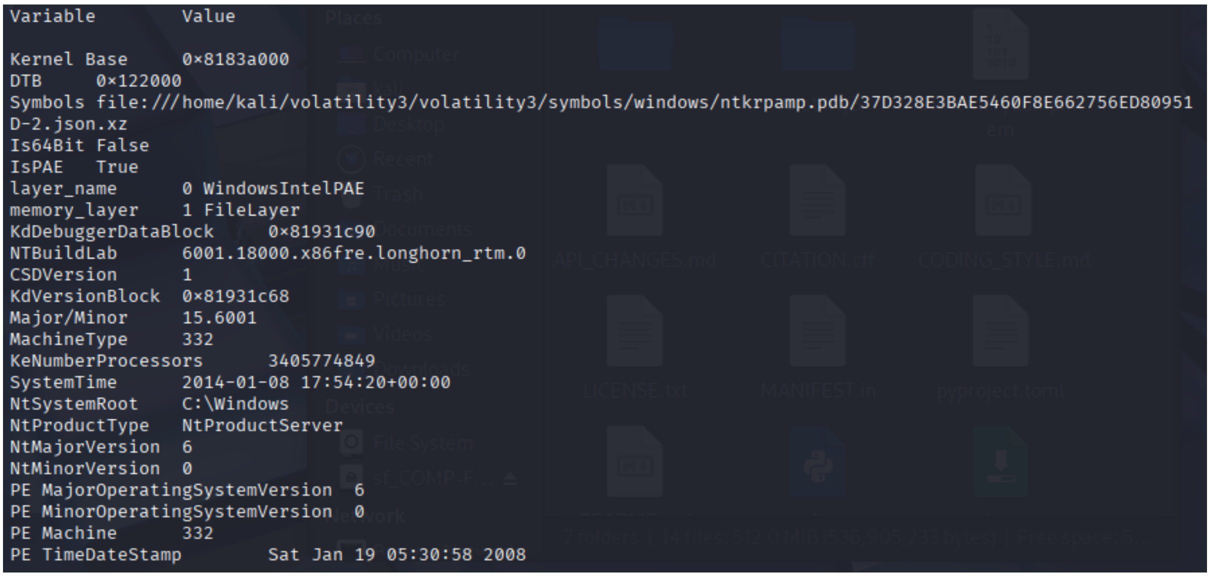
a. Memory dump's basic information.

The figures below showcase the command to acquire the memory dump's basic information along with its output.



```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.info
```

Figure 47: Command to show memory dunp's basic information



```
Variable        Value

Kernel Base     0x8183a000
DTB     0x122000
Symbols file:///home/kali/volatility3/volatility3/symbols/windows/ntkrpamp.pdb/37D328E3BAE5460F8E662756ED80951D-2.json.xz
Is64Bit False
IsPAE   True
layer_name      0 WindowsIntelPAE
memory_layer    1 FileLayer
KdDebuggerDataBlock     0x81931c90
NTBuildLab      6001.18000.x86fre.longhorn_rtm.0
CSDVersion      1
KdVersionBlock  0x81931c68
Major/Minor     15.6001
MachineType     332
KeNumberProcessors      3405774849
SystemTime      2014-01-08 17:54:20+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductServer
NtMajorVersion  6
NtMinorVersion  0
PE MajorOperatingSystemVersion  6
PE MinorOperatingSystemVersion  0
PE Machine      332
PE TimeDateStamp        Sat Jan 19 05:30:58 2008
```

Figure 48: Memory dump's basic information

b. List of running processes

Figure 49 showcases the command to acquire and print the list of running processes into a text file named “pslist.txt”. Additionally, Figure 50 showcases the contents of the file.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.pslist > pslist.txt
```

Figure 49: Command to show list of running processes

```
Volatility 3 Framework 2.26.2

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output

4	0	System	0x82db0910	100	541	N/A	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
404	4	smss.exe	0x8454c118	4	28	N/A	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
472	460	csrss.exe	0x84561968	11	466	0	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
516	508	csrss.exe	0x84450770	10	305	1	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
524	460	wininit.exe	0x84453770	3	98	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
552	508	winlogon.exe	0x84465770	3	116	1	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
604	524	services.exe	0x83632170	6	250	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
616	524	lsass.exe	0x844bf770	13	610	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
624	524	lsm.exe	0x844c2680	10	208	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
788	604	svchost.exe	0x84866d50	6	298	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
848	604	svchost.exe	0x845f37a8	8	280	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
884	604	svchost.exe	0x848fa118	15	274	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
976	604	svchost.exe	0x84914d90	6	152	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1000	604	svchost.exe	0x8491bd90	45	2072	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1056	604	SLsvc.exe	0x8492a6d0	4	96	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1088	604	svchost.exe	0x84937d90	17	567	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1160	604	svchost.exe	0x84941d90	20	265	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1188	604	svchost.exe	0x84945c30	22	596	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1308	604	svchost.exe	0x8496e9f0	17	265	0	False	2014-01-08 02:17:43.000000 UTC	N/A	Disabled
1424	604	spoolsv.exe	0x849c18a8	17	291	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1460	604	armsvc.exe	0x849d7610	2	56	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1480	604	dns.exe	0x849dcd90	10	164	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1508	604	ftpbasicsvr.exe	0x849e1cc0	2	52	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1576	604	svchost.exe	0x849f5888	5	124	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1604	604	svchost.exe	0x849faad8	3	73	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1620	604	snmp.exe	0x849f7380	4	147	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1640	604	vmtoolsd.exe	0x84a0b020	7	273	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1696	604	svchost.exe	0x84a21d90	4	44	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
832	604	TPAutoConnSvc.e	0x83066b68	9	136	0	False	2014-01-08 02:17:52.000000 UTC	N/A	Disabled
1924	604	dllhost.exe	0x84ba8d90	13	240	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
1572	604	msdtc.exe	0x84bb7ca8	11	167	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
2096	1000	taskeng.exe	0x84ba1c30	5	137	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
2352	1000	taskeng.exe	0x84c13020	10	250	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2368	552	userinit.exe	0x84c148e8	0	-	1	False	2014-01-08 02:18:17.000000 UTC	2014-01-08 02:18:43.000000 UTC	Disabled
2392	1160	dwm.exe	0x84c1d020	3	76	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2480	832	TPAutoConnect.e	0x84c2c898	2	103	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2496	2368	explorer.exe	0x84c2c020	24	689	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2580	2496	vmtoolsd.exe	0x84c5e020	6	9004	1	False	2014-01-08 02:18:18.000000 UTC	N/A	Disabled
2592	2496	AdobeARM.exe	0x84c5f020	6	282	1	False	2014-01-08 02:18:18.000000 UTC	N/A	Disabled
2616	2592	reader_sl.exe	0x84c537b0	0	-	1	False	2014-01-08 02:18:18.000000 UTC	2014-01-08 02:19:20.000000 UTC	Disabled
2720	2444	Oobe.exe	0x84c11298	0	-	1	False	2014-01-08 02:18:22.000000 UTC	2014-01-08 02:55:43.000000 UTC	Disabled
3224	604	svchost.exe	0x84c73b60	9	228	0	False	2014-01-08 02:19:53.000000 UTC	N/A	Disabled
3336	788	iashost.exe	0x84ce3020	2	97	0	False	2014-01-08 02:19:53.000000 UTC	N/A	Disabled
3680	1000	wuauclt.exe	0x84bd8020	2	139	1	False	2014-01-08 02:20:55.000000 UTC	N/A	Disabled
3920	2496	notepad.exe	0x84c64a50	1	51	1	False	2014-01-08 03:19:07.000000 UTC	N/A	Disabled
1800	2496	FTK Imager.exe	0x84cfd958	5	251	1	False	2014-01-08 03:19:32.000000 UTC	N/A	Disabled
1888	2496	iexplore.exe	0x848ab618	14	641	1	False	2014-01-08 03:20:24.000000 UTC	N/A	Disabled
2708	2496	notepad.exe	0x848a1340	1	45	1	False	2014-01-08 17:33:08.000000 UTC	N/A	Disabled
```

Figure 50: Memory dump's list of running processes

c. List of processes with detailed information

Figure 51 showcases the command to gather the list of processes with detailed information and print the output into a file named “psscan.txt”. Figure 52 showcase the contents of the text file.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.psscan > psscan.txt
```

Figure 51: Command to show processes with detailed information

```
Volatility 3 Framework 2.26.2

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output

788	604	svchost.exe	0x c4ad50	7	298	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
2616	2592	reader_sl.exe	0xc817b0	0	-	1	False	2014-01-08 02:18:18.000000 UTC	2014-01-08 02:19:20.000000 UTC	Disabled
4	0	System	0x2fb0910	100	541	N/A	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
832	604	TPAutoConnSvc.e	0x4ff9b68	9	136	0	False	2014-01-08 02:17:52.000000 UTC	N/A	Disabled
832	604	TPAutoConnSvc.e	0x7f68b68	9	136	0	False	2014-01-08 02:17:52.000000 UTC	N/A	Disabled
832	604	TPAutoConnSvc.e	0x886bb68	9	136	0	False	2014-01-08 02:17:52.000000 UTC	N/A	Disabled
2720	2444	Oobe.exe	0x1e011298	0	-	1	False	2014-01-08 02:18:22.000000 UTC	2014-01-08 02:55:43.000000 UTC	Disabled
2352	1000	taskeng.exe	0x1e013020	10	250	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2368	552	userinit.exe	0x1e0148e8	0	-	1	False	2014-01-08 02:18:17.000000 UTC	2014-01-08 02:18:43.000000 UTC	Disabled
2392	1160	dwm.exe	0x1e01d020	3	76	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2496	2368	explorer.exe	0x1e02c020	24	689	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2480	832	TPAutoConnect.e	0x1e02c898	2	103	1	False	2014-01-08 02:18:17.000000 UTC	N/A	Disabled
2616	2592	reader_sl.exe	0x1e0537b0	0	-	1	False	2014-01-08 02:18:18.000000 UTC	2014-01-08 02:19:20.000000 UTC	Disabled
2580	2496	vmtoolsd.exe	0x1e05e020	6	9004	1	False	2014-01-08 02:18:18.000000 UTC	N/A	Disabled
2592	2496	AdobeARM.exe	0x1e05f020	6	282	1	False	2014-01-08 02:18:18.000000 UTC	N/A	Disabled
3920	2496	notepad.exe	0x1e064a50	1	51	1	False	2014-01-08 03:19:07.000000 UTC	N/A	Disabled
3224	604	svchost.exe	0x1e073b60	9	228	0	False	2014-01-08 02:19:53.000000 UTC	N/A	Disabled
3336	788	iashost.exe	0x1e0e3020	2	97	0	False	2014-01-08 02:19:53.000000 UTC	N/A	Disabled
1800	2496	FTK Imager.exe	0x1e0fd958	5	251	1	False	2014-01-08 03:19:32.000000 UTC	N/A	Disabled
1640	604	vmtoolsd.exe	0x1e20b020	7	273	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1696	604	svchost.exe	0x1e221d90	4	44	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
2096	1000	taskeng.exe	0x1e3a1c30	5	137	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
1924	604	dllhost.exe	0x1e3a8d90	13	240	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
1572	604	msdtc.exe	0x1e3b7ca8	11	167	0	False	2014-01-08 02:17:53.000000 UTC	N/A	Disabled
3680	1000	wuauclt.exe	0x1e3d8020	2	139	1	False	2014-01-08 02:20:55.000000 UTC	N/A	Disabled
788	604	svchost.exe	0x1e466d50	6	298	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
2708	2496	notepad.exe	0x1e4a1340	1	45	1	False	2014-01-08 17:33:08.000000 UTC	N/A	Disabled
1888	2496	iexplore.exe	0x1e4ab618	14	641	1	False	2014-01-08 03:20:24.000000 UTC	N/A	Disabled
884	604	svchost.exe	0x1e4fa118	15	274	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
976	604	svchost.exe	0x1e514d90	6	152	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1000	604	svchost.exe	0x1e51bd90	45	2072	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1056	604	SLsvc.exe	0x1e52a6d0	4	96	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1088	604	svchost.exe	0x1e537d90	17	567	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1160	604	svchost.exe	0x1e541d90	20	265	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1188	604	svchost.exe	0x1e545c30	22	596	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
1308	604	svchost.exe	0x1e56e9f0	17	265	0	False	2014-01-08 02:17:43.000000 UTC	N/A	Disabled
1424	604	spoolsv.exe	0x1e5c18a8	17	291	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1460	604	armsvc.exe	0x1e5d7610	2	56	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1480	604	dns.exe	0x1e5dcd90	10	164	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1508	604	ftpbasicsvr.exe	0x1e5e1cc0	2	52	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1576	604	svchost.exe	0x1e5f5888	5	124	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1620	604	snmp.exe	0x1e5f7380	4	147	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
1604	604	svchost.exe	0x1e5faad8	3	73	0	False	2014-01-08 02:17:49.000000 UTC	N/A	Disabled
516	508	csrss.exe	0x1e850770	10	305	1	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
524	460	wininit.exe	0x1e853770	3	98	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
552	508	winlogon.exe	0x1e865770	3	116	1	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
616	524	lsass.exe	0x1e8bf770	13	610	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
624	524	lsm.exe	0x1e8c2680	10	208	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
404	4	smss.exe	0x1e94c118	4	28	N/A	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
472	460	csrss.exe	0x1e961968	11	466	0	False	2014-01-08 02:17:35.000000 UTC	N/A	Disabled
848	604	svchost.exe	0x1e9f37a8	8	280	0	False	2014-01-08 02:17:42.000000 UTC	N/A	Disabled
604	524	services.exe	0x1f632170	6	250	0	False	2014-01-08 02:17:36.000000 UTC	N/A	Disabled
832	604	TPAutoConnSvc.e	0x1fc66b68	9	136	0	False	2014-01-08 02:17:52.000000 UTC	N/A	Disabled
```

Figure 52: List of processes with detailed information

d. List of handles of processes

Figure 53 showcase the command to get the list of handles of processes and write them to a text file named “handles.txt”. Figure 54 showcases the contents of the file.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.handles > handles.txt
```

Figure 53: Command to show list of handles

```
Volatility 3 Framework 2.26.2

PID     Process Offset  HandleValue     Type    GrantedAccess   Name

4       System  0x82db0910      0x4     Process 0x1fffff        System Pid 4
4       System  0x86213398      0x8     Key     0x2001f MACHINE\SYSTEM\CONTROLSET001\CONTROL\HIVELIST
4       System  0x8620aa90      0xc     Key     0x0     -
4       System  0x8625e770      0x10    Key     0x2001f MACHINE\SYSTEM\CONTROLSET001\CONTROL\PRODUCTOPTIONS
4       System  0x8625cd68      0x14    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\SESSION MANAGER\MEMORY MANAGEMENT\PREFETCHPARAMETERS
4       System  0x87a4dad0      0x18    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x86261188      0x1c    Key     0x2001f MACHINE\SYSTEM\SETUP
4       System  0x862d7138      0x20    Key     0x20019 MACHINE\SYSTEM\CONTROLSET001\CONTROL\WMI\SECURITY
4       System  0x86246210      0x24    Key     0x20019 MACHINE\HARDWARE\DESCRIPTION\SYSTEM\MULTIFUNCTIONADAPTER
4       System  0x82ee33b0      0x28    Thread  0x1fffff        Tid 184 Pid 4
4       System  0x83ee14e0      0x2c    Thread  0x0     Tid 304 Pid 4
4       System  0x88406f18      0x30    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\INITVERBS
4       System  0x86368020      0x34    Token   0xe     -
4       System  0x8842d938      0x38    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x863ba830      0x3c    Key     0x20019 MACHINE\HARDWARE\DESCRIPTION\SYSTEM\BIOS
4       System  0x88419690      0x40    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x862d5020      0x44    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x884011d0      0x48    Key     0x20019 MACHINE\SYSTEM\CONTROLSET001\SERVICES\DISK
4       System  0x87a52480      0x4c    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x8625d3a8      0x50    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x88423eb0      0x54    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x8633a850      0x58    Key     0xf003f MACHINE\HARDWARE\DEVICEMAP\SCSI\SCSI PORT 3\SCSI BUS 0
4       System  0x836b1b40      0x5c    File    0x120116        \Device\Mup
4       System  0x8841c6f0      0x60    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x8362da08      0x64    TmRm    0x12007f        -
4       System  0x8362a810      0x68    TmTm    0xf003f -
4       System  0x8362d028      0x6c    TmRm    0x1f007f        -
4       System  0x8362a6b0      0x70    File    0x12019f        \Device\clfsKtmLog
4       System  0x83629b78      0x74    File    0x13019f        \Device\clfs\Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLog
4       System  0x83623d90      0x78    File    0x12019f        \Device\clfs\Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLog
4       System  0x835e0408      0x7c    File    0x1     \Device\HarddiskVolume1
4       System  0x8314a600      0x80    File    0x13019f        \Device\clfs\Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLog
4       System  0x835e0870      0x84    File    0x12019f        \Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLogContainer00000000000000000002
4       System  0x835e0918      0x88    File    0x12019f        \Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLogContainer00000000000000000001
4       System  0x831672e0      0x8c    File    0x12019f        \Device\HarddiskVolume1\$Extend\$RmMetadata\$TxfLog\$TxfLog.blf
4       System  0x83629530      0x90    Thread  0x1fffff        Tid 232 Pid 4
4       System  0x8372aa68      0x94    File    0x12019f        \Device\clfsTxfLog
4       System  0x8625a350      0x98    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x86202670      0x9c    Token   0xf01ff -
4       System  0x82db0910      0xa0    Process 0x1440  System Pid 4
4       System  0x863b5518      0xa4    Key     0x20019 MACHINE\SYSTEM\CONTROLSET001
4       System  0x8633a8a0      0xa8    Key     0x10    MACHINE\SYSTEM\CONTROLSET001\CONTROL\NETWORKPROVIDER\ORDER
4       System  0x862bfa78      0xac    Key     0x20019 MACHINE\SYSTEM\CONTROLSET001\SERVICES\DISK
4       System  0x863f57d8      0xb0    Key     0x20019 MACHINE\SYSTEM\CONTROLSET001\SERVICES\MUP
4       System  0x830a4780      0xb4    Thread  0x1f0fff        Tid 200 Pid 4
4       System  0x86252e18      0xb8    Key     0xf003f MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4D36E96C-E325-11CE-BFC1-08002BE10318}\0000\MIXERSETTINGS
4       System  0x86251f50      0xbc    Key     0xf003f MACHINE\HARDWARE\DEVICEMAP\SCSI\SCSI PORT 2\SCSI BUS 0
```

Figure 54: List of handles

e. Handles of the process with PID=1056

Figure 55 showcases the command to show the list of handles of the process with PID 1056, along with the output of the command.

```
└─$ python3 vol.py -f a1memorydump.mem windows.handles --pid 1056
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
PID     Process Offset  HandleValue     Type    GrantedAccess   Name

1056    SLsvc.exe       0×89cd9948      0×4     Directory       0×3     KnownDlls
1056    SLsvc.exe       0×8492c340      0×8     File    0×100020        \Device\HarddiskVolume1\Windows\System
32
1056    SLsvc.exe       0×8492a438      0×c     Event   0×1f0003        -
1056    SLsvc.exe       0×8492a3f8      0×10    Mutant  0×1f0001        -
1056    SLsvc.exe       0×8492b4a8      0×14    ALPC Port       0×1f0001        -
1056    SLsvc.exe       0×863ad318      0×18    Key     0×20019 MACHINE
1056    SLsvc.exe       0×8492d098      0×1c    Event   0×1f0003        -
1056    SLsvc.exe       0×862d06e0      0×20    Key     0×1     MACHINE\SYSTEM\CONTROLSET001\CONTROL\SESSION M
ANAGER
1056    SLsvc.exe       0×845f69f0      0×24    WindowStation   0×f006e Service-0×0-3e4$
1056    SLsvc.exe       0×8492d310      0×28    Event   0×21f0003       -
1056    SLsvc.exe       0×845f6930      0×2c    Desktop 0×f00cf Default
1056    SLsvc.exe       0×845f69f0      0×30    WindowStation   0×f006e Service-0×0-3e4$
1056    SLsvc.exe       0×849310c0      0×34    EtwRegistration 0×804   -
1056    SLsvc.exe       0×8492da08      0×38    Event   0×1f0003        -
1056    SLsvc.exe       0×84931068      0×3c    EtwRegistration 0×804   -
1056    SLsvc.exe       0×80ca3cc0      0×40    Token   0×48    -
1056    SLsvc.exe       0×84932e98      0×44    File    0×12019f        \Device\NamedPipe\net\NtControlPipe6
1056    SLsvc.exe       0×89d99330      0×48    Directory       0×f     BaseNamedObjects
1056    SLsvc.exe       0×84932d80      0×4c    Event   0×1f0003        -
1056    SLsvc.exe       0×84932d50      0×50    Event   0×1f0003        -
1056    SLsvc.exe       0×84932d20      0×54    Event   0×1f0003        -
1056    SLsvc.exe       0×84932cd0      0×58    EtwRegistration 0×804   -
1056    SLsvc.exe       0×84932c98      0×5c    Event   0×1f0003        -
1056    SLsvc.exe       0×8492c030      0×60    Thread  0×1fffff        Tid 1060 Pid 1056
1056    SLsvc.exe       0×84932b80      0×64    ALPC Port       0×1f0001        -
1056    SLsvc.exe       0×849338c8      0×70    Event   0×1f0003        -
1056    SLsvc.exe       0×84934728      0×74    Event   0×1f0003        -
1056    SLsvc.exe       0×849328a8      0×78    Event   0×1f0003        -
1056    SLsvc.exe       0×8492e758      0×7c    Timer   0×1f0003        -
1056    SLsvc.exe       0×84933620      0×80    Thread  0×1fffff        Tid 1084 Pid 1056
1056    SLsvc.exe       0×84933620      0×84    Thread  0×1fffff        Tid 1084 Pid 1056
1056    SLsvc.exe       0×913be800      0×88    KeyedEvent      0×f0003 -
1056    SLsvc.exe       0×84934020      0×8c    IoCompletion    0×1f0003        -
1056    SLsvc.exe       0×84934510      0×90    TpWorkerFactory 0×f00ff -
1056    SLsvc.exe       0×88431158      0×94    Section 0×f0007 DSEC420
1056    SLsvc.exe       0×84937bd0      0×98    ALPC Port       0×1f0001        SLCTransportEndpoint-00001
1056    SLsvc.exe       0×84937b70      0×9c    IoCompletion    0×1f0003        -
1056    SLsvc.exe       0×84937b20      0×a0    IoCompletion    0×1f0003        -
1056    SLsvc.exe       0×84937b70      0×a4    IoCompletion    0×1f0003        -
1056    SLsvc.exe       0×845e93e0      0×a8    Event   0×1f0003        -
1056    SLsvc.exe       0×84939ac0      0×ac    Thread  0×1fffff        Tid 1100 Pid 1056
1056    SLsvc.exe       0×8493b020      0×b0    ALPC Port       0×1f0001        -
1056    SLsvc.exe       0×9138e078      0×b4    KeyedEvent      0×f0003 -
1056    SLsvc.exe       0×8491a020      0×b8    Timer   0×100002        -
1056    SLsvc.exe       0×84938228      0×bc    Timer   0×100002        -
1056    SLsvc.exe       0×8495c870      0×c0    EtwRegistration 0×804   -
```

Figure 55: Handles of process with PID=1056

f. List of hidden processes

Figure 56 showcases the command to acquire the list of hidden processes and write the output into a file named “hidden.txt”. Figure 57 shows the contents of the file.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.psxview > hidden.txt
```

Figure 56: Command to show list of hidden processes

```
Volatility 3 Framework 2.26.2

Offset(Virtual) Name    PID     pslist  psscan  thrdscan        csrss   Exit Time

0x849d7610      armsvc.exe      1460    True    False   True    True
0x1e5d7610      armsvc.exe      1460    False   True    False   False
0x848ab618      iexplore.exe    1888    True    False   True    True
0x1e4ab618      iexplore.exe    1888    False   True    False   False
0x1e3d8020      wuauclt.exe     3680    False   True    False   False
0x1e01d020      dwm.exe 2392    False   True    False   False
0x84a0b020      vmtoolsd.exe    1640    True    False   True    True
0x84c13020      taskeng.exe     2352    True    False   True    True
0x84c1d020      dwm.exe 2392    True    False   True    True
0x84c2c020      explorer.exe    2496    True    False   True    True
0x84c5e020      vmtoolsd.exe    2580    True    False   True    True
0x84c5f020      AdobeARM.exe    2592    True    False   True    True
0x84ce3020      iashost.exe     3336    True    False   True    True
0x84bd8020      wuauclt.exe     3680    True    False   True    True
0x1e013020      taskeng.exe     2352    False   True    False   False
0x1e05f020      AdobeARM.exe    2592    False   True    False   False
0x84945c30      svchost.exe     1188    True    False   True    True
0x84ba1c30      taskeng.exe     2096    True    False   True    True
0x1e3a1c30      taskeng.exe     2096    False   True    False   False
0x1e545c30      svchost.exe     1188    False   True    False   False
0x1e0e3020      iashost.exe     3336    False   True    False   False
0x1e5dcd90      dns.exe 1480    False   True    False   False
0x84c64a50      notepad.exe     3920    True    False   True    True
0x1e064a50      notepad.exe     3920    False   True    False   False
0x844c2680      lsm.exe 624     True    False   True    True
0x1e8c2680      lsm.exe 624     False   True    False   False
0x849f5888      svchost.exe     1576    True    False   True    True
0x1e5f5888      svchost.exe     1576    False   True    False   False
0x84c2c898      TPAutoConnect.e 2480    True    False   True    True
0x84c11298      Oobe.exe        2720    True    False   False   False   2014-01-08 02:55:43+00:00
0x1e011298      Oobe.exe        2720    False   True    False   False   2014-01-08 02:55:43+00:00
0x1e02c898      TPAutoConnect.e 2480    False   True    False   False
0x1e02c020      explorer.exe    2496    False   True    False   False
0x849c18a8      spoolsv.exe     1424    True    False   True    True
0x84bb7ca8      msdtc.exe       1572    True    False   True    True
0x1e3b7ca8      msdtc.exe       1572    False   True    False   False
0x1e5c18a8      spoolsv.exe     1424    False   True    False   False
0x1e541d90      svchost.exe     1160    False   True    False   False
0x849e1cc0      ftpbasicsvr.exe 1508    True    False   True    True
0x1e5e1cc0      ftpbasicsvr.exe 1508    False   True    False   False
0x8492a6d0      SLsvc.exe       1056    True    False   True    True
```

Figure 57: List of hidden processes

g. Registry hives

Figure 58 shows the command to acquire the registry hives, along with the output of the command.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.registry.hivelist
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Offset  FileFullPath    File output

0x812eb6b0      \Device\HarddiskVolume1\Windows\ServiceProfiles\NetworkService\NTUSER.DAT       Disabled
0x81321008      \Device\HarddiskVolume1\Windows\ServiceProfiles\LocalService\NTUSER.DAT Disabled
0x86211008              Disabled
0x86226008      \REGISTRY\MACHINE\SYSTEM        Disabled
0x86248008      \REGISTRY\MACHINE\HARDWARE      Disabled
0x89c2f148      \Device\HarddiskVolume1\Windows\System32\config\DEFAULT Disabled
0x89c33450      \Device\HarddiskVolume1\Windows\System32\config\SAM     Disabled
0x89c36008      \Device\HarddiskVolume1\Windows\System32\config\SECURITY        Disabled
0x89c47008      \Device\HarddiskVolume1\Windows\System32\config\COMPONENTS      Disabled
0x89c47a20      \Device\HarddiskVolume1\Windows\System32\config\SOFTWARE        Disabled
0x89cd1a20      \Device\HarddiskVolume1\Boot\BCD        Disabled
0x9465f6a8      \Device\HarddiskVolume1\Users\Administrator\NTUSER.DAT  Disabled
0x946ae008      \Device\HarddiskVolume1\Users\Administrator\AppData\Local\Microsoft\Windows\UsrClass.dat        Disabled
```

Figure 58: Registry hives

h. List of Dynamic Link Libraries (DLLs)

Figure 59 shows the command to acquire the list of DLLs and write the list to a text file named “dlllist.txt”. Figure 60 showcases the contents of the file.

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.dlllist > dlllist.txt
```

Figure 59: Command to show DLLs

```
Volatility 3 Framework 2.26.2

PID     Process Base    Size    Name    Path    LoadTime        File output

404     smss.exe        0x47610000      0x12000 -       -       N/A     Disabled
404     smss.exe        0x77d00000      0x127000        -       -       N/A     Disabled
472     csrss.exe       0x4a130000      0x5000  csrss.exe       C:\Windows\system32\csrss.exe   N/A     Disabled
472     csrss.exe       0x77d00000      0x127000        -       -       N/A     Disabled
472     csrss.exe       0x764d0000      0xf000  CSRSRV.dll      C:\Windows\system32\CSRSRV.dll  N/A     Disabled
472     csrss.exe       0x764b0000      0x13000 basesrv.dll     C:\Windows\system32\basesrv.dll N/A     Disabled
472     csrss.exe       0x76450000      0x60000 winsrv.dll      C:\Windows\system32\winsrv.dll  N/A     Disabled
472     csrss.exe       0x765a0000      0x9d000 USER32.dll      C:\Windows\system32\USER32.dll  N/A     Disabled
472     csrss.exe       0x76bb0000      0xdb000 KERNEL32.dll    C:\Windows\system32\KERNEL32.dll        N/A     Disabled
472     csrss.exe       0x77f10000      0x4b000 GDI32.dll       C:\Windows\system32\GDI32.dll   N/A     Disabled
472     csrss.exe       0x779d0000      0xc6000 ADVAPI32.dll    C:\Windows\system32\ADVAPI32.dll        N/A     Disabled
472     csrss.exe       0x77c20000      0xc3000 RPCRT4.dll      C:\Windows\system32\RPCRT4.dll  N/A     Disabled
472     csrss.exe       0x76580000      0x9000  LPK.DLL C:\Windows\system32\LPK.DLL     N/A     Disabled
472     csrss.exe       0x77b50000      0x7d000 USP10.dll       C:\Windows\system32\USP10.dll   N/A     Disabled
472     csrss.exe       0x77e60000      0xaa000 msvcrt.dll      C:\Windows\system32\msvcrt.dll  N/A     Disabled
472     csrss.exe       0x762c0000      0x5f000 sxs.dll C:\Windows\system32\sxs.dll     N/A     Disabled
516     csrss.exe       0x4a130000      0x5000  csrss.exe       C:\Windows\system32\csrss.exe   N/A     Disabled
516     csrss.exe       0x77d00000      0x127000        -       -       N/A     Disabled
516     csrss.exe       0x764d0000      0xf000  CSRSRV.dll      C:\Windows\system32\CSRSRV.dll  N/A     Disabled
516     csrss.exe       0x764b0000      0x13000 basesrv.dll     C:\Windows\system32\basesrv.dll N/A     Disabled
516     csrss.exe       0x76450000      0x60000 winsrv.dll      C:\Windows\system32\winsrv.dll  N/A     Disabled
516     csrss.exe       0x765a0000      0x9d000 USER32.dll      C:\Windows\system32\USER32.dll  N/A     Disabled
516     csrss.exe       0x76bb0000      0xdb000 KERNEL32.dll    C:\Windows\system32\KERNEL32.dll        N/A     Disabled
516     csrss.exe       0x77f10000      0x4b000 GDI32.dll       C:\Windows\system32\GDI32.dll   N/A     Disabled
516     csrss.exe       0x779d0000      0xc6000 ADVAPI32.dll    C:\Windows\system32\ADVAPI32.dll        N/A     Disabled
516     csrss.exe       0x77c20000      0xc3000 RPCRT4.dll      C:\Windows\system32\RPCRT4.dll  N/A     Disabled
516     csrss.exe       0x76580000      0x9000  LPK.DLL C:\Windows\system32\LPK.DLL     N/A     Disabled
516     csrss.exe       0x77b50000      0x7d000 USP10.dll       C:\Windows\system32\USP10.dll   N/A     Disabled
516     csrss.exe       0x77e60000      0xaa000 msvcrt.dll      C:\Windows\system32\msvcrt.dll  N/A     Disabled
516     csrss.exe       0x762c0000      0x5f000 sxs.dll C:\Windows\system32\sxs.dll     N/A     Disabled
604     services.exe    0x710000        0x47000 services.exe    C:\Windows\system32\services.exe        N/A     Disabled
604     services.exe    0x77d00000      0x127000        ntdll.dll       C:\Windows\system32\ntdll.dll   N/A     Disabled
604     services.exe    0x76bb0000      0xdb000 kernel32.dll    C:\Windows\system32\kernel32.dll        N/A     Disabled
604     services.exe    0x779d0000      0xc6000 ADVAPI32.dll    C:\Windows\system32\ADVAPI32.dll        N/A     Disabled
604     services.exe    0x77c20000      0xc3000 RPCRT4.dll      C:\Windows\system32\RPCRT4.dll  N/A     Disabled
604     services.exe    0x765a0000      0x9d000 USER32.dll      C:\Windows\system32\USER32.dll  N/A     Disabled
604     services.exe    0x77f10000      0x4b000 GDI32.dll       C:\Windows\system32\GDI32.dll   N/A     Disabled
604     services.exe    0x77e60000      0xaa000 msvcrt.dll      C:\Windows\system32\msvcrt.dll  N/A     Disabled
604     services.exe    0x76410000      0x1e000 USERENV.dll     C:\Windows\system32\USERENV.dll N/A     Disabled
604     services.exe    0x76430000      0x14000 Secur32.dll     C:\Windows\system32\Secur32.dll N/A     Disabled
604     services.exe    0x76330000      0x4e000 SCESRV.dll      C:\Windows\system32\SCESRV.dll  N/A     Disabled
```

Figure 60: List of DLLs

i. List of DLLs that are being used by PID 2708

Figure 61 showcases the command to get the list of DLLs being used by the process with PID 2708 and write the output of the command into a csv file named "dll_pid2708". Figure 62 showcases the contents of the file on the terminal via the command "cat dll_pid2708.csv".

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f a1memorydump.mem windows.dlllist --pid 2708 > dll_pid2708.csv
```

Figure 61: Command to show list of DLLs being used by PID=2078

```
Volatility 3 Framework 2.26.2

PID     Process Base    Size    Name    Path    LoadTime        File output

2708    notepad.exe     0x b90000       0x28000 notepad.exe     C:\Windows\System32\notepad.exe N/A     Disabled
2708    notepad.exe     0x77d00000      0x127000        ntdll.dll       C:\Windows\system32\ntdll.dll   N/A     Disabled
2708    notepad.exe     0x76bb0000      0xdb000 kernel32.dll    C:\Windows\system32\kernel32.dll        N/A     Disabled
2708    notepad.exe     0x779d0000      0xc6000 ADVAPI32.dll    C:\Windows\system32\ADVAPI32.dll        N/A     Disabled
2708    notepad.exe     0x77c20000      0xc3000 RPCRT4.dll      C:\Windows\system32\RPCRT4.dll  N/A     Disabled
2708    notepad.exe     0x77f10000      0x4b000 GDI32.dll       C:\Windows\system32\GDI32.dll   N/A     Disabled
2708    notepad.exe     0x765a0000      0x9d000 USER32.dll      C:\Windows\system32\USER32.dll  N/A     Disabled
2708    notepad.exe     0x77e60000      0xaa000 msvcrt.dll      C:\Windows\system32\msvcrt.dll  N/A     Disabled
2708    notepad.exe     0x76870000      0x73000 COMDLG32.dll    C:\Windows\system32\COMDLG32.dll        N/A     Disabled
2708    notepad.exe     0x76cc0000      0x58000 SHLWAPI.dll     C:\Windows\system32\SHLWAPI.dll N/A     Disabled
2708    notepad.exe     0x75150000      0x19e000        COMCTL32.dll    C:\Windows\WinSxS\x86_microsoft.windows.common-controls_65
2708    notepad.exe     0x76e70000      0xb0f000        SHELL32.dll     C:\Windows\system32\SHELL32.dll N/A     Disabled
2708    notepad.exe     0x73780000      0x42000 WINSPOOL.DRV    C:\Windows\System32\WINSPOOL.DRV        N/A     Disabled
2708    notepad.exe     0x76d20000      0x144000        ole32.dll       C:\Windows\system32\ole32.dll   N/A     Disabled
2708    notepad.exe     0x767e0000      0x8d000 OLEAUT32.dll    C:\Windows\system32\OLEAUT32.dll        N/A     Disabled
2708    notepad.exe     0x77b30000      0x1e000 IMM32.DLL       C:\Windows\system32\IMM32.DLL   N/A     Disabled
2708    notepad.exe     0x76640000      0xc8000 MSCTF.dll       C:\Windows\system32\MSCTF.dll   N/A     Disabled
2708    notepad.exe     0x76580000      0x9000  LPK.DLL C:\Windows\system32\LPK.DLL     N/A     Disabled
2708    notepad.exe     0x77b50000      0x7d000 USP10.dll       C:\Windows\system32\USP10.dll   N/A     Disabled
2708    notepad.exe     0x756a0000      0x3f000 UxTheme.dll     C:\Windows\System32\UxTheme.dll N/A     Disabled
```

Figure 62: List of DLLs being used by PID=2078 - shown in csv

```
┌──(kali㉿kali)-[~/volatility3]
└─$ cat dll_pid2708.csv
Volatility 3 Framework 2.26.2

PID     Process Base    Size    Name    Path    LoadTime        File output

2708    notepad.exe     0xb90000        0x28000 notepad.exe     C:\Windows\System32\notepad.exe N/A     Disabled
2708    notepad.exe     0x77d00000      0x127000        ntdll.dll       C:\Windows\system32\ntdll.dll   N/A     Disabled
2708    notepad.exe     0x76bb0000      0xdb000 kernel32.dll    C:\Windows\system32\kernel32.dll        N/A     Disabled
2708    notepad.exe     0x779d0000      0xc6000 ADVAPI32.dll    C:\Windows\system32\ADVAPI32.dll        N/A     Disabled
2708    notepad.exe     0x77c20000      0xc3000 RPCRT4.dll      C:\Windows\system32\RPCRT4.dll  N/A     Disabled
2708    notepad.exe     0x77f10000      0x4b000 GDI32.dll       C:\Windows\system32\GDI32.dll   N/A     Disabled
2708    notepad.exe     0x765a0000      0x9d000 USER32.dll      C:\Windows\system32\USER32.dll  N/A     Disabled
2708    notepad.exe     0x77e60000      0xaa000 msvcrt.dll      C:\Windows\system32\msvcrt.dll  N/A     Disabled
2708    notepad.exe     0x76870000      0x73000 COMDLG32.dll    C:\Windows\system32\COMDLG32.dll        N/A     Disabled
2708    notepad.exe     0x76cc0000      0x58000 SHLWAPI.dll     C:\Windows\system32\SHLWAPI.dll N/A     Disabled
2708    notepad.exe     0x75150000      0x19e000        COMCTL32.dll    C:\Windows\WinSxS\x86_microsoft.windows.common-cor
2708    notepad.exe     0x76e70000      0xb0f000        SHELL32.dll     C:\Windows\system32\SHELL32.dll N/A     Disabled
2708    notepad.exe     0x73780000      0x42000 WINSPOOL.DRV    C:\Windows\System32\WINSPOOL.DRV        N/A     Disabled
2708    notepad.exe     0x76d20000      0x144000        ole32.dll       C:\Windows\system32\ole32.dll   N/A     Disabled
2708    notepad.exe     0x767e0000      0x8d000 OLEAUT32.dll    C:\Windows\system32\OLEAUT32.dll        N/A     Disabled
2708    notepad.exe     0x77b30000      0x1e000 IMM32.DLL       C:\Windows\system32\IMM32.DLL   N/A     Disabled
2708    notepad.exe     0x76640000      0xc8000 MSCTF.dll       C:\Windows\system32\MSCTF.dll   N/A     Disabled
2708    notepad.exe     0x76580000      0x9000  LPK.DLL C:\Windows\system32\LPK.DLL     N/A     Disabled
2708    notepad.exe     0x77b50000      0x7d000 USP10.dll       C:\Windows\system32\USP10.dll   N/A     Disabled
2708    notepad.exe     0x756a0000      0x3f000 UxTheme.dll     C:\Windows\System32\UxTheme.dll N/A     Disabled
```

Figure 63: List of DLLs being used by PID=2078 - shown in terminal